DIVISION OF NAVOTAS CITY

Senior High School

BUSINESS MATHEMATICS

(Quarter 1 and 2)
First Semester

ADM
ALTERNATIVE DELIVERY MODE

S.Y. 2021-2022

NAVOTAS CITY PHILIPPINES

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Business Mathematics for Senior High School
Alternative Delivery Mode
1st Semester (Quarter 1&2)
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What I Know

Directions: Choose the letter of the correct answer. Write your answer on a separate sheet of paper.

Quarter I

1. Three out of four children in a school love chocolate products. Represent the fraction of children who love chocolates to decimal form.
   A. 0.34 C. 3
   B. 0.75 D. 3.4
2. Ramon is 68 years old. As a senior citizen, he is exempted to VAT and is also entitled for a 20% discount on food items. If he orders a meal and the cost of the food is Php 90.00, how much will she pay the cashier?
   A. Php 28.80 C. Php 64.29
   B. Php 61.20 D. Php 72.00
3. The more the workers there are, the lesser the number of hours is needed to complete a task. This is an example of what type of proportion?
   A. direct proportion C. partitive proportion
   B. inverse proportion D. mixed proportion

For numbers 4-5: 3 pieces of lumpiang sariwa cost Php 120. How much will it cost for 15 pieces?

4. How will you translate this problem into a proportion?
   A. $\frac{3}{120} = \frac{15}{n}$ C. $\frac{3}{15} = \frac{n}{120}$
   B. $\frac{3}{120} = \frac{n}{15}$ D. $\frac{3}{n} = \frac{15}{120}$
5. How much will be the cost if you buy 15 pieces of lumpiang sariwa?
   A. Php 240.00 C. Php 600.00
   B. Php 360.00 D. Php 720.00
6. Rose would like to sell mobile phones she purchased from a supplier. The cost of each phone is Php 3,500.00 each. The operating cost is set at 15% of the cost and she would like to have a 25% profit on the cost of each item. How much should be the selling price?
   A. Php 3,525.00 C. Php 4,500.00
   B. Php 3,900.00 D. Php 4,900.00
7. PQR Company sells laptops worth ₱24,999 each. Each unit costs 75% of its selling price. What is the rate of margin?
   A. 23.5% B. 25% C. 28.7% D. 75%
8. What is the single equivalent rate if the merchant gives you a trade discount series of 20/10/10?
   A. 43.5% B. 40% C. 35% D. 35.2%
9. How much is the profit/loss of a business if the revenue is ₱120,500 and the expenses is Php102,000?
   A. Profit; Php 18,500.00 C. Loss; Php 18,500.00
   B. Profit; Php 120,500.00 D. Loss; Php 102,000.00

Quarter II

For numbers 10-11: You are planning to buy a condominium unit worth ₱2,000,000. However, your savings is not enough to fully paid the property, so you applied for a mortgage from a bank. The bank required you to make a 10% down payment.

10. How much is the down payment?
    A. Php 200,000.00
    B. Php 20,000.00
    C. Php 2,000.00
    D. Php 200.00
11. What is the mortgage loan?
    A. Php 2,200,000.00
    B. Php 2,000,000.00
    C. Php 1,800,000.00
    D. Php 1,600,000.00
12. Adolfo works in a company that pays Php 2,000 a week and a 12% commission on sales above Php 15,000 in the week. How much will he receive if he sold products worth Php 28,000 in a week.
    A. Php 15,000.00
    B. Php 13,000.00
    C. Php 4,650.00
    D. Php 3,560.00
13. A car dealer earns a 2.5% commission on sales. If he sells a car for Php 980,000, how much commission will he earn?
    A. Php 24,500.00
    B. Php 25,000.00
    C. Php 245,000.00
    D. Php 2,450,000.00
14. A worker earns a salary of Php 36,000 a month. Suppose that he works 8 hours a day for a 24-day work month, how much does he get per hour?
    A. Php 150.00
    B. Php 187.50
    C. Php 1,500.00
    D. Php 36,000.00
15. Which of the following benefits is taxable?
    A. SSS contribution
    B. Benefits over Php 90,000
    C. Rice subsidy amounting to Php 2,000
    D. 13th month pay not exceeding to Php 90,000
16. An employee works as a clerk in ATL Company, a private lending company in a barangay. His monthly gross income is Php 25,000. How much is the employee's share to Pag-IBIG Fund?
    A. Php 100.00
    B. Php 150.00
    C. Php 200.00
    D. Php 250.00
17. A wage earner who earns Php 537.00 a day rendered ten hours in work during a regular day. How much pay will he receive that day?
    A. Php 83.91
    B. Php 167.81
    C. Php 704.81
    D. Php 939.75
18. A wage earner who earns Php 537.00 a day reported to work during a regular holiday. How much pay will he receive that day?
    A. Php 671.25
    B. Php 805.50
    C. Php 939.75
    D. Php 1,074.00
19. In entering a formula inside a cell, the first symbol you must write is ___.
    A. >
    B. <
    C. =
    D. ^
20. Which chart is appropriate to use if you want to show trends over time?
    A. Line Graph
    B. Bar Chart
    C. Pie Chart
    D. Tables

MODULE1

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Business Mathematics. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

After going through this module, you are expected to:

1. Express fractions to decimals and percent forms, decimals to fraction and percent forms and precent to fractions and decimal forms; and
2. Give real-life situations to illustrate fractions, decimals, and percent.

Lesson 1 Fractions, Decimals, Percent

You always encounter fractions, decimals, and percent even when you don't notice it. Whether you like it or not, they have been part of your daily living, and your understanding of these concepts helps you in making better decisions in life. For instance, you use these ideas in managing your personal finances, buying decisions, and many more. In addition to its essence in practical life, expertise on these concepts is necessary and foundational to more complex concepts in Math.

In this lesson, you will go back to the basics. Your mastery in these topics will greatly help you in your next learning experience.

What is It

In your daily living, you experienced instances where it is better to use fractions than decimals, and vice-versa. Thus, learning how to convert fractions, decimals, or percent to other forms will give you an edge in doing daily estimation and calculation.

Fractions

Expressing fractions to decimals

There are two ways to convert fractions to decimals.

1. By using long division

   Example: Express 5/8 to decimal form.
   Solution: The numerator is the dividend and the denominator is the divisor.

```
        .625
8 ) 5.000
   -48
     20
    -16
      40
     -40
       0
```

   Therefore, 5/8 can also be rewritten as 0.625

2. By converting the fraction to an equivalent fraction whose denominator is a power of 10.

   Example: Convert 3/4 to decimal form.

STEPS	SOLUTION
Step 1: Think of a number that can be multiplied by the denominator to make it 10, 100, 1000, 10 000, and so on (power of 10).	The denominator is 4, so you can multiply it by 25 to make the denominator as a power of 10.
Step 2: Multiply the numerator and the denominator by that number.	$\frac{3}{4} \cdot \frac{25}{25} = \frac{75}{100}$
Step 3: Write down the numerator, then move the decimal point to the left depending on the number of zeros in the denominator.	In this case, there are two zeros in the denominator, so move the decimal point two places to the left. Therefore, the answer is 0.75.

Decimals

Expressing decimals to fractions.

1. When the decimal is terminating, use the concept of place value.

(The place value for decimals starts with tenths, hundredths, thousandths, and so on.)

Examples:

a. *Convert 0.8 to fraction.*
   Solution: 0.8 is read as "8 tenths"

$$0.8 = \frac{8}{10} = \frac{4}{5}$$

b. *Express 0.25 to fraction form.*
   Solution: 0.25 is read as "25 hundredths"

$$0.25 = \frac{25}{100} = \frac{1}{4}$$

2. When the decimal is repeating and non-terminating, do the following:

Examples:

a. *Write 0.3333… to fraction.* (0.333… can also be rewritten as $0.\overline{3}$)

STEPS	SOLUTIONS
Let x = $0.\overline{3}$ (first equation)	The decimal has one repeating decimal place so multiply both sides by 10. The resulting equation will be 10x = 3.3333…
10x = 3.3333… (second equation) -x = 0.3333… 9x = 3	Subtract the first equation from the second equation.
$\frac{9x}{9} = \frac{3}{9} = \frac{1}{3}$	Solve for x.

b. *Express 0.454545… to fraction.*

STEPS	SOLUTIONS
Let x = 0.454545 ... (1st equation)	The decimal has two repeating decimal places so multiply both sides by 100. The resulting equation will be 100x = 45.454545…
100x =45.454545 ... (2nd equation) -x = 0.454545… 99x = 45	Subtract the first equation from the second equation.
$\frac{99x}{99} = \frac{45}{99} = \frac{5}{11}$	Solve for x.

Note: Always write your answer in its simplest form.

Percent

Percent means "per hundred." It demonstrates a fraction whose denominator is 100.

Wondering about the difference between "percent" and "percentage"? According to Peterson (2008), the only difference between these two words depends on how you used them. Percentage speaks of the concept whereas percent serves as a unit. For example, you do not say "50 percentage." Instead, you say "50% or 50 percent." The word percentage does not come with a number while percent always come with a number.

Example: The percentage of ABM students who passed in the previous semester was 95%.

A. Expressing decimal to percent
Just multiply the decimals by 100 (or simply move the decimal point two places to the right), and then add % symbol.

Examples:
1. 0.125 = 12.5% 2. 0.08 = 8% 3. 1.3 = 130%

B. Expressing percent to decimal
Divide the percent by 100%, or simply move the decimal point two places to the left. Then, omit the % symbol.

Examples:
1. 64% = 0.64 2. 125% = 1.25 3. 1.5% = 0.015

C. Expressing fraction to percent
First, convert the fraction to decimal. Then, convert the resulting decimal to percent.

Examples:
1. $\frac{1}{5} = 0.2 = 20\%$ 2. $\frac{5}{8} = 0.625 = 62.5\%$ 3. $\frac{9}{5} = 1.8 = 180\%$

What's More

Activity: The Mystery Guy

Mimi is a saleslady in a supermarket. One time after her noon break, Lily, her co-saleslady, handed her a bunch of flowers and a chocolate bar. According to Lily, these were given by one of their supervisors in the branch. Mimi wanted to find out who was the mystery guy. The only clue given to her was that the guy is very good at Math.

There are only four supervisors in the branch where Mimi works. Help her find the mystery guy by examining the given mathematical statements inside the box. Remember that the mystery guy is very good at math, so he would be able to give all correct answers.

Determine whether the statement is True or False.

1. $\frac{7}{8} = 0.875$	6. 4.25 = 425%
2. 0.555… = $\frac{5}{9}$	7. $\frac{3}{5} = 0.6$
3. 0.456 = 45.6	8. $\frac{5}{7} = \frac{7}{5}$
4. 15% = 0.15	9. 6% = 0.6
5. $\frac{6}{8}$ = 68%	10. 0.0025 = 0.25%

Supervisor #1

No. 1 is True

No. 3 is True

No. 5 is True

Supervisor #2

No. 2 is True

No. 7 is True

No. 10 is False

Supervisor #3

No. 3 is False

No. 5 is False

No. 10 is True

Supervisor #4

No. 4 is True

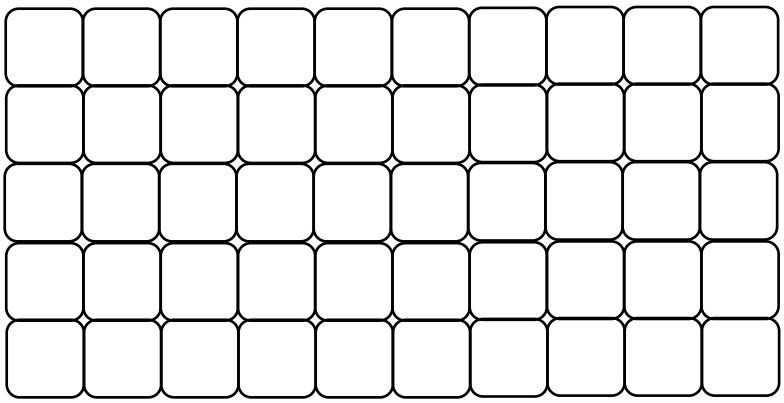
No. 6 is True

No. 9 is True

Who is the mystery guy? _____

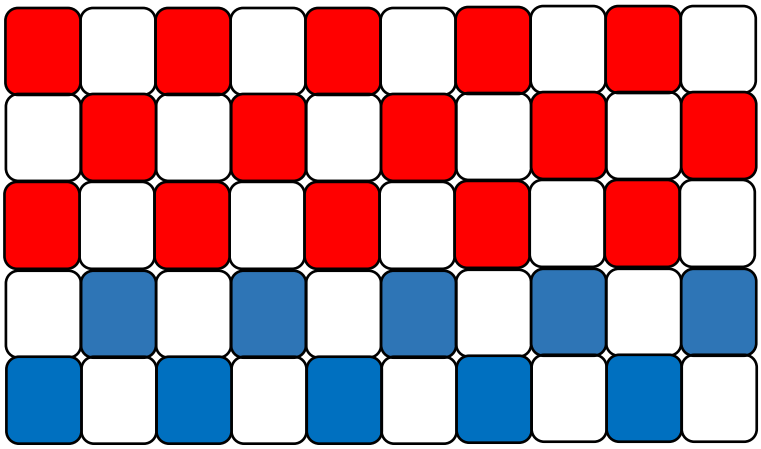
What I Can Do

Let's Do Quilting!



1. Look at the mat above. Then, copy this on a bond paper.
2. Make the copy as large as your bond paper so you can work on it.
3. Assume that you will be selling this mat online. However, it must be unique and attractive so buyers would be interested to buy.
4. Beautify the item by coloring the box using your preferred colors.
5. Finally, make a table showing the components of your item.

Example:



Components based on the given example:

Color	Fraction	Decimal	Percent
Red	$\frac{15}{50} = \frac{3}{10}$	0.3	30%
Blue	$\frac{10}{50} = \frac{1}{5}$	0.2	20%
White	$\frac{25}{50} = \frac{1}{2}$	0.5	50%
TOTAL	1	1	100%

RUBRIC:

	4	3	2	1	YOUR SCORE
Neatness	The finished product is neat and without any defects like stains, folds, rips.	The finished product is neat, but with minor defect/s.	The finished product has a minor issue on cleanliness and defects.	The finished product shows major defects and is obviously unclean.	
Creativity	The finished product is well-planned, unique and interesting.	The finished product is not so unique, but well-planned and interesting.	The finished product is common, but interesting.	The finished product is not unique and not well planned.	
Mathematical Ideas	The table of data contains no error.	The table of data contains 1 to 2 errors.	The table of data contains 3 to 4 errors	The table of data contains more than 4 errors.	
Requirement	All directions were followed completely.	One of the directions was not followed completely.	Two of the directions were not followed completely.	More than 2 instructions were not followed completely.	
Timeliness	The output was submitted on time.	The output was submitted 1-2 days late.	The output was submitted 3-4 days late	The output was submitted more than 4 days late.	
TOTAL SCORE					

MODULE 2

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Business Mathematics. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of learners. The lessons are arranged to follow the standard sequence of the course.

The module is divided into two lessons, namely:

- Lesson 2.1 – Operations on Fractions, Decimals, and Percent
- Lesson 2.2 – Problems Involving Fractions, Decimals, and Percent

After going through this module, you are expected to:

1. convert between fractions, decimals, and percent if needed in the problem.
2. determine which operation(s) on fractions, decimals and/or percent must be used to solve problems involving fractions, decimals and/or percent; and
3. solve problems involving fractions, decimals, and percent.

Lesson 2.1 Fractions, Decimals, Percent

What is It

Business always involves analysis and computation. Hence, you must constantly practice the basic skills you've learned in mathematics. For this module, you will start the lesson by studying how to perform the operations involving fractions, decimals, and percent.

A. Operations on Fraction

1. Addition/Subtraction of Fractions
   a. Similar Fractions (Fractions with the same denominator)
      - *Rule*: $\frac{a}{c}+\frac{b}{c}=\frac{a+b}{c}$, $c \neq 0$

      Example: $\frac{1}{12}+\frac{6}{12}=\frac{7}{12}$ When fractions are similar, just add/subtract the numerators and copy the denominator.

   b. Dissimilar Fractions (Fractions with different denominators)
      - Find the LCM and convert the fractions to similar fractions.
      - Then, do the steps in adding/subtracting similar fractions.

      Example: $\frac{1}{4}+\frac{3}{10}$

Step	Solution
Find the LCM of 4 and 10	$4 = 2 \cdot 2$ $\underline{10 = 2 \cdot 5}$ $LCM = 2 \cdot 2 \cdot 5 = 20$
Use the LCM in converting fractions to similar fractions, then add.	$\frac{1}{4}+\frac{3}{10}=\frac{5+6}{20}=\frac{11}{20}$
Note: Always simplify your final answer.	

2. Multiplication of Fractions
   Multiply the numerator by the other numerator, and the denominator by the other denominator. Do not cross multiply.
   Rule: $\frac{a}{b}\cdot\frac{c}{d}=\frac{ac}{bd}$, $b,d \neq 0$

   Example: $\frac{2}{3}\cdot\frac{1}{5}=\frac{2}{15}$ This equation is also read as $\frac{2}{3}$ *of* $\frac{1}{5}$ *is* $\frac{2}{15}$.

   Note: The preposition "***of***" signifies multiplication.
   Example: $\frac{3}{4}$ *of* 20 means $\frac{3}{4}\cdot\frac{20}{1}=\frac{60}{4}=15$.

3. Division of Fractions
   Either you get the reciprocal of the divisor and proceed to the rule in multiplication, or just do cross multiplication.

   Rule: $\frac{a}{b}\div\frac{c}{d}=\frac{a}{b}\cdot\frac{d}{c}=\frac{ad}{bc}$

   Example: $\frac{3}{4}\div\frac{1}{10}=\frac{3}{4}\cdot\frac{10}{1}=\frac{30}{4}=\frac{15}{2}$

B. Operations on Decimals

1. Addition/Subtraction of Decimals
   Steps: Align the decimal points. Then add/subtract as if you are adding or subtracting whole numbers. Finally, bring down the decimal point.
   Example 1: Add 115.25, 30 and 4.27.
   115.25

+ 30.00 Whole number has its decimal points after the ones digit place.

```
      4.27
    149.52
```

Example 2: Subtract 14.75 from 25.
Solution: Take note of the word "*from.*"

```
  25.00
- 14.75
  10.25
```

2. Multiplication of Decimals
   Steps: Multiply the decimals as if you are multiplying whole numbers. The total number of decimal places of the factors is also the number of decimal places in the product.

   Example: Multiply 2.256 by 1.8.

   Solution:

```
      2.256      (3 decimal places)
  ×     1.8      (1 decimal place)
    18048
  +  2256
    4.0608       (4 decimal places)
```

3. Division of Decimals
   There are three cases in dividing decimals.

   Case 1: Both the divisor and dividend are whole numbers, but the quotient has a fractional part

   - In this case, divide the decimals just like in whole numbers, then put the decimal point directly above the decimal point in the dividend.

   Example: 15 ÷ 4.
   Solution:

```
      3.75
4 )15.00
   - 12
      30
   -  28
       20
    -  20
        0
```

Case 2: The divisor is a whole number and the dividend has a fractional part

- Put the decimal point of the quotient directly above the decimal point in the dividend.

Example: 37.2 ÷ 6

Solution:

```
      6.2
```

```
6 )37.2
   -36
     12
   -  12
      0
```

Case 3: The divisor has a fractional part.

- Multiply both the divisor and the dividend by an appropriate multiple of 10 to make the divisor a whole number, then apply Case 2.

Example: 4.64 ÷ 0.8

Steps	Solution
The divisor is 0.8. To make it a whole number, multiply it by 10. Multiply also the dividend by 4.64. (*Clue: Count the number of digits after the decimal point. In this example, there is only one digit after the point, so just multiply it by 10. If there are 2 digits after the point, then multiply it by 100.*)	0.8 (10) = 8 4.64 (10) = 46.4 *Recall that a shorter way to multiply a decimal by* 10^n *is by moving the decimal point n places to the right.*
Apply Case 2	6.2 8)46.4 -40 64 - 64 0

C. Operations involving Percent

When performing an operation involving percent, you have to

1. convert percent to decimal; and
2. apply the rule in multiplying decimals.

Example: Find the product of 75% and Php 240.
Solution:

```
     240
x     .75
    1200
+ 1680_
  180.00
```

As a Business Math student, you have to be familiar with all the problems that may involve fractions, decimals, and percent. Before you proceed to the next example, you have to be familiar with the three quantities you will be working with:

- Base (original amount) – is the whole in a problem.

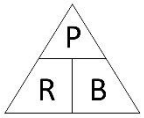
- Percentage (or amount) – is the portion of the whole being compared to the base.
- Rate (or percent) – is the ratio of the percentage to base. It is written as percent.

Practice identifying these:

	Base (B)	**Rate (R)**	**Percentage (P)**
20% of 50 is 10	50	20%	10
7.5 is 30% of 25	25	30%	7.5
18 is 75% of 24	24	75%	18

Hint: The base comes after the word "of"
The rate is the one with % symbol

To solve for the unknown, you just have to remember this triangle.



- P = R * B
- B = $\frac{P}{B}$
- R = $\frac{P}{B}$

Example 1: What is 30% of 12?

Given	Solution
B = 12 R = 30% or 0.3 P = ?	P = R * B P = .3 * 12 P = 3.6

Example 2: 25% of what number is 50?

Given	Solution
P = 50 R = 25% or 0.25 B = ?	B = $\frac{P}{B} = \frac{50}{.25} = 200$

Example 3: 10 is what percent of 40?

Given	Solution
P = 10 R = ? B = 40	R = $\frac{P}{B} = \frac{10}{40}$ R = 0.25 Convert to 0.25 to percent R = 25%

Lesson 2.2 Problems Involving Fractions, Decimals and Percent

Solving word problems require reading comprehension. Even if you are good in mathematical concepts, you can still go wrong if you do not understand the problems well. In this module, you will encounter several word problems, so read everything carefully.

What is It

Solving Problems Involving Fractions, Decimals, Percent

Every day you solve different word problems, not just in school, but also in other areas of your life. So, whether you like it or not, it is a necessity that you know what to do when you encounter them. Schoenborn & Simkins (n.d.) suggested the following tips in solving any word problems. Use only those that apply to the situation.

1. Read the problem carefully. Do not proceed to any step until you understand the problem well.
2. Find and list the given.
3. Identify what is being asked in the problem.
4. Eliminate information that are not necessary to find the solution.
5. Be attentive to the units of measure. There are times that you need to convert units first before solving the problem.
6. If needed, draw a diagram to visualize the problem.
7. Identify or develop a formula to use for the computation.

Sample Problem 1: Aling Nena goes to a market to buy $3\frac{1}{2}$ kg pork, $\frac{3}{4}$ kg of chicken, and 3 L of cooking oil. The following is the list of prices:

Description	Amount per Unit
Pork	Php 280.00
Chicken	Php 210.00
Cooking Oil	Php 25.75

Questions:

1. How much is the total amount of all the items bought by Aling Nena?
   Solution:

   For pork : $3\frac{1}{2}\ (280) = 3.5\ (280) = 980.00$

For chicken : $\frac{3}{4}$ (210) = 0.75 (210) = 157.50
For cooking oil : 3 (25.75) = 77.25
Total : 980 + 157.50 + 77.25 = 1,214.75

The total amount of the items is Php 1, 214.75.

2. If Aling Nena has Php 1500, how much will be left to her after paying all the items?
   Solution:

$$\begin{array}{r} 1500.00 \\ -\ 1214.75 \\ \hline 285.25 \end{array}$$

The money left to her is Php 285.25.

Note: Always indicate the peso sign at the final answer if it pertains to money.

Sample Problem 2: Your parents instructed you to call your grandma in the province. If your current load balance is Php 145.00 and a call would cost you Php 6.50 per minute (*assuming that you are not registered to any call and text promo*), how many minutes will you be able to talk with your grandma?

Solution:

145 ÷ 6.5 = 22.31

You will be able to call your grandma for 22 minutes.

Sample Problem 3: A laptop costs Php 23 500 before the 12% VAT. How much is the total cost of the laptop after the VAT?

Solution:

- Php 23 500 has no VAT yet. We call this as VAT-able amount.
- Multiply the VAT-able amount by 112%.
  - 112% = 1.12
  - 23 500 (1.12) = 26 320
  - The total cost of laptop after VAT is Php 26 320.00

Sample Problem 4: Your total water bill is Php 321.50. How much of it is the VAT-able amount?

Solution:

- Take note of the following:
  - Php 321.50 is already inclusive of VAT.
  - The amount of VAT is 12% of the VAT-able amount.

Step	Solution
Write a mathematical statement for this.	Let x = VAT-able amount So, o x + 12% of x = 321.50 o x + .12x = 321.50
Solve for x	1.12 x = 321.50

	$\frac{1.12x}{1.12} = \frac{321.50}{1.12}$ x ≈ 287.054
Final answer	The VAT-able amount or the price that is VAT free is Php 287.05

Sample Problem 5: Linda is 65 years old. As a senior citizen, she is exempted to VAT and is also entitled for a 20% discount on food items. If she orders a meal and the cost of the food is Php 90.00, how much will she pay the cashier?

Solution:

- Take note that the 20% discount is applied to VAT-able amount. Therefore, compute first for the VAT-able amount.
  - $x = \frac{90}{1.12} = 80.36$
- So, what is 80% of 80.36? (Linda will only pay for 80% of the amount because she is given with 20% discount.)
  - 80.36*0.8 = 64.29

Answer: Linda's exact bill for her food is Php 64.29.

Activity: Your Turn to Perform!

Refer to the following data to answer the problems below. Round your answers to the nearest hundredths (2 decimal places).

Product	Amount per Unit
Notebook	Php 12.50
Pad paper	Php 9.25
Ballpen	Php 6.50
Pencil	Php 6.00
Paste	Php 4.75
Eraser	Php 2.50
Crayon	Php 32.75

You need to buy school supplies for the new school year. However, you do not want to hustle with people who are also buying in the store. So, you decided to contact someone to deliver it for you. The delivery fee is 25% of the total amount of purchase.

Here are your selected items:

12 pcs of notebook
2 pcs. pad paper
3 pcs. ballpen
1 paste
1 box of crayon

1. How much is the total amount of items you intend to buy?

2. How much will you pay for the delivery charge?

3. Because you buy a dozen of notebook, the store gives you 8% discount for the notebooks you bought. How much will be your total bill without the delivery charge? With the delivery charge?

4. The amount you see in the table are inclusive of VAT. Find the VAT-able amount of your purchased product.

5. You have a 20% discount voucher, and you want to use that in ordering your supplies. How much will you pay (without the delivery charge) if you use the said voucher?

What I Can Do

Activity: Valuing Your Values

This activity contains some values you should possess in solving problems not only in Math but also in daily life. Find it out by following the given instructions.

Instructions:

1. Perform the indicated operations.
2. Always follow the PEMDAS (***P***arenthesis, ***E***xponent, ***M***ultiplication, ***D***ivision, ***A***ddition and ***S***ubtraction) Rule.
3. Round the decimals to the nearest hundredths.
4. Write the fraction to its simplest form.
5. Then, use the result to figure out the hidden word in each bullet.

A = $\frac{2}{9} + \frac{1}{9}$

E = $\frac{4}{5} + \frac{1}{8}$

I = $\frac{17}{20} - \frac{1}{2}$

O = $\frac{9}{16}\left(\frac{1}{3}\right)$

U = $\frac{4}{7} \div \frac{2}{3}$

C = 0.43+9.5 - 3.61

D = 14.75 – 5.25 + 13

F = 90 ÷ 0.15

G = 2+ (0.3) (1.48)

L = 19.84 + 4.5 – 2

N = 18% of 1500

P = 25% of 6800

R = 12.5% of 350

S = 15% of $\frac{2}{5}$

T = 60% of $\frac{1}{5}$

- _____ _____ _____ _____
  2.444 43.75 $\frac{7}{20}$ 0.12

- _____ _____ _____ _____ _____
  600 $\frac{9}{48}$ 6.32 $\frac{6}{7}$ 0.06

- _____ _____ _____ _____ _____ _____ _____ _____
  1700 $\frac{1}{3}$ 0.12 $\frac{7}{20}$ $\frac{37}{40}$ 270 6.32 $\frac{37}{40}$

- _____ _____ _____ _____ _____ _____ _____ _____ _____
  22.5 $\frac{7}{20}$ 22.34 $\frac{7}{20}$ 2.444 $\frac{37}{40}$ 270 6.32 $\frac{37}{40}$

MODULE 3

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Business Mathematics. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of learners. The lessons are arranged to follow the standard sequence of the course.

After going through this module, you are expected to:

1. define and give examples of a ratio;
2. express a ratio as a fraction in simplest forms;
3. simplify a ratio involving rational numbers or decimals;
4. formulate and solve problems involving ratios;
5. differentiate between ratio and rate;
6. define and give examples of a proportion;
7. write proportions involving real-life situations;
8. use fundamental property of proportions to show if two ratios form a proportion;
9. use fundamental property of proportions to solve a proportion; and
10. identify the different kinds of proportions and write examples of real-life situations for each.

Lesson 3	Ratio and Proportion

Do you know that the terms "ratio" and "fraction" are used interchangeably? That is because problems involving fraction can also be considered as ratio problems too. This means that just like fractions, you also encounter ratios in real-life. In this lesson, you will learn more about this topic. You will notice that the way you handle fraction is also the same as the way you handle ratio.

What is It

Comparing things happens every day and everywhere. When you go to the market, you make comparisons before buying a product. At school, you also make comparisons when you check your score over a total number of items. Even in the comfort of your house, you make comparisons when you decide how much rice will you cook with the number of people who will eat. Even if you are not aware of it, ratio and proportion apply to your daily living.

A. Ratio

The comparison of two quantities by division is called ratio, and it can be written in two ways:

Odds Notation	Fraction Notation
a:b read as "***a*** *is to* ***b***" or "***a*** *to* ***b***"	$\frac{a}{b}$

Example 1: The ratio of 6 to 8 can be written as any of these:

- 6:8 or 3:4 (odds notation)
- $\frac{6}{8}$ *or* $\frac{3}{4}$ (fractional notation)

Example 2: There are 15 boys and 10 girls in a classroom. What is the ratio of girls to boys?

Solution:

- Note that you are asked for the ratio of ***girls*** to ***boys***.

- There are 10 girls and 15 boys or $\frac{10}{15}$.
- In simplest form, we can write $\frac{10}{15}$ to $\frac{2}{3}$.
- So, the ratio of girls to boys is $\frac{2}{3}$ or 2 : 3.

Example 3: Write the following ratios in its simplest form.

a. $\frac{1}{2} : \frac{3}{5}$

Steps	Solution
Find the LCM of the denominator.	The denominators are 2 and 5. The LCM is 10.
Multiply both quantities by the LCM.	$10\left(\frac{1}{2}\right) : 10\left(\frac{3}{5}\right)$ $\frac{10}{2} : \frac{30}{5} = 5:6$
Write the simplest form.	5:6

b. $1\frac{2}{3} : \frac{3}{4}$

Steps	Solution
Rewrite $1\frac{2}{3}$ to improper fraction.	$1\frac{2}{3} = \frac{5}{3}$ Convert the mixed number to improper fraction $\frac{5}{3} : \frac{3}{4}$
Find the LCM of the denominator.	The denominators are 3 and 4. The LCM is 12.
Multiply both quantities by the LCM.	$12\left(\frac{5}{3}\right) : 12\left(\frac{3}{4}\right)$ $\frac{60}{3} : \frac{36}{4} = 20:9$
Write the simplest form.	20:9

c. 0.04 : 0.06

Steps	Solution
Write the ratio to its equivalent form without decimals by multiplying ***both*** terms by the ***same*** power of 10.	You need to multiply 0.04 and 0.06 by 100 to make both terms a whole number. $\frac{0.04\ (100)}{0.06\ (100)} = \frac{4}{6}$
Write to the ratio to its simplest form.	$\frac{4}{6} = \frac{2}{3}$ Thus, the simplest form is 2 : 3.

d. 0.25 : 0.675

Steps	Solution
Write the ratio to its equivalent form without decimals by multiplying ***both*** terms by the ***same*** power of 10.	You need to multiply 0.25 and 0.675 by 1000 to make both terms a whole number. $\frac{0.25\,(1000)}{0.675\,(1000)} = \frac{250}{675} = \frac{10}{27}$
Write to the ratio to its simplest form.	Thus, the simplest form is 10 : 27.

Before solving any problems involving ratio, make sure that both quantities or terms are expressed in the same unit of measure. Look at the next examples.

Example 4: Express the ratio of 1500 meters to 2 kilometers to simplest form.

> Convert one of the terms to make them of the same unit of measure.
> Know that 1 kilometer = 1000 meter
> Convert 2 kilometers to meters. So, instead of using 2 km, use 2000 m.

- 1500 meters to 2000 meters
- 1500 : 2000 or $\frac{1500}{2000} = \frac{3}{4}$
- The simplest form is 3 : 4

Will it also be the same if instead of converting kilometers to meters, you choose to convert meters to kilometers? Let's see.

Steps	Solution
Convert 1500 meters to kilometers.	$1500\ \cancel{m} = \frac{1\ km}{1000\ \cancel{m}}$ $\frac{1500\ km}{1000} = 1.5\ km$
Write the equivalent ratio with the same unit of measure.	1.5 km to 2 km or 1.5 : 2
Note that 1.5 is in decimals so you have to make it a whole number.	$\frac{1.5(10)}{2(10)} = \frac{15}{20} = \frac{3}{4}$
Make a statement.	The simplest form is 3 : 4

Example 5: Write the ratio, 32 days to 10 weeks, as a fraction in simplest forms.

> Convert one of the terms to make them of the same unit of measure.
> Know that 1 week = 7 days
> Convert 10 weeks to days. So, instead of using 10 weeks, use 70 days.

- 32 days to 70 days

- 32 : 70 or $\frac{32}{70} = \frac{16}{35}$
- The simplest form is 16 : 35

B. Rate

Rate is a ratio. Meaning, it also compares two quantities. However, the term "rate" is used to compare quantities of different units. When the rate has a denominator of 1, it is called a ***unit rate.***

Example 1: Write the unit rate for 400 km in 8 hours.

(Note that in unit rate, your goal is to make the denominator equal to 1.)

$$\frac{400\ km}{8\ hr} = \frac{50\ km}{1\ hr}\ or\ 50\ km/hr$$

Example 2: Give the unit rate when a pump moves 50 liters in 4 minutes.

$$\frac{50\ liters}{4\ minutes} = \frac{12.5\ liters}{1\ minute}\ or\ 12.5\ liters/minute$$

C. Proportion

Proportion is a statement that the two ratios are equal. According to the fundamental property of proportions, the two given ratios are equal when the product of the means is equal to the product of the extremes. In symbols,

$$if\ \frac{a}{b} = \frac{c}{d}\ , then\ ad = bc$$

where **a** is the first term, **b** is the second term, **c** is the third term, and **d** is the fourth term. a and d are called the **extremes** and b and c are called the **means**.

Example 1: Name the four terms in the following proportion:

$\frac{2}{5} = \frac{10}{25}$ or 2 : 5 = 10 : 25

Solution:

First Term = 2 Second Term = 5 Third Term = 10 Fourth Term = 25

2 and 25 are the means; 5 and 10 are the extremes.

Example 2: Do the following ratios form a proportion?

a. 3 : 8 $\stackrel{?}{=}$ 4 : 9

Solution:

27 **32**

$\frac{3}{8}$ $\frac{4}{9}$ 27 ≠ 32 Hence, the ratios do not form a proportion.

b. $\frac{20}{25} \overset{?}{=} \frac{4}{5}$

Solution:

100 **100**

$\frac{20}{25} \times \frac{4}{5}$ 100 = 100 Hence, the ratios form a proportion.

Example 3: Find the missing term in the following proportions.

a. 10 : 15 = n : 18

Solution:

Steps	**Solution**
Rewrite the ratio in fraction form then get the cross products	$\frac{10}{15} = \frac{n}{18}$ 15n = 10(18) 15n = 180
Solve for n	$\frac{15n}{15} = \frac{180}{15}$ n = 12

Will it also be the same if you simplify first $\frac{10}{15}$ *to* $\frac{2}{3}$ *before solving for n? Let's try!*

Solution:

$\frac{2}{3} = \frac{n}{18}$ → 3n = 2(18) → 3n = 36 → $\frac{3n}{3} = \frac{36}{3}$ → n = 12

b. n : 15 = 1 : 5

Solution:

$\frac{n}{15} = \frac{1}{5}$ → 5n = 15(1) → 5n = 15 → $\frac{5n}{5} = \frac{15}{5}$ → n = 3

The relationship of variables can be either they are directly related, inversely related or partitively related. Recognizing the type of variation will help you solve different problems involving ratio and proportion.

Three Types of Proportion

1. **Direct Proportion**
   The two variables are both increasing or both decreasing at the same rate.

   Example:
   If you work 15 hours, you earn Php 600. How much will you earn if you work 60 hours?

   - Note that as the number of hours you work increases; the amount you earn also increases. This is a type of direct proportion.

2. **Inverse Proportion**
   The proportion is inverse if as one quantity increases, the other quantity decreases at the same rate, and vice versa.

   Example:
   If 3 men can paint a wall in 1 day, how long will 5 men do it?

   - This is an indirect or inverse proportion because the more workers, the lesser the time they will finish the painting.

3. **Partitive Proportion**
   A whole is divided into several parts given a ratio.

   Examples:
   Fifty liters of water is transferred into 3 jars in the ratio 2:3:4. How much water is in each jar?

   - This is an example of partitive proportion because the whole 50 liters is divided into three parts in the ratio 2:3:5.

What's More

Activity: The Secret Ingredient!

Maria is problematic because her profit in bread and pastry business is declining due to the competition in the market. Suddenly, she remembered that when she was a child, her grandma was telling her about a secret ingredient in making a cake. However, this was kept inside a box with a code.

Help Maria open the box and find the secret ingredient by solving for ***n*** in the following proportions.

C	$\frac{3}{4} = \frac{12}{n}$	H	$\frac{6}{n} = \frac{18}{27}$	N	$\frac{n}{8} = \frac{21}{24}$
D	$\frac{9}{15} = \frac{n}{5}$	I	$\frac{n}{7} = \frac{16}{56}$	O	$\frac{10}{20} = \frac{n}{2}$
E	$\frac{n}{10} = \frac{16}{20}$	L	$\frac{4}{18} = \frac{n}{54}$	R	$\frac{n}{5} = \frac{16}{20}$
S	$40 : n = 80 : 10$	T	$24 : 12 = n : 3$		
V	$30 : 45 = 10 : n$				

			6	9	8		5	8	16	4	8	6					
2	7		G	4	8	3	2	8	7	6			2	5			
						12	1	15	8								

What I Can Do

This is now your time to apply the concept you have learned about ratio and proportion. Suppose you will make a delicious pastillas.

Ingredients:

- 1 ¼ cups of powdered milk
- ¼ cup condensed milk
- ¼ cup sugar
- Oil, for molding

These ingredients can yield 30 small pieces of pastillas.

Instructions:

1. You will give 3 pieces of pastillas to each member of your family.
2. Since the ingredients above yield 30 pieces, you may reduce or increase the amount based on the number of people who will eat the pastillas.
3. Complete the table.

Number of people who will receive pastillas:	
Number of pastillas needed to make:	
Ingredients	**Actual Amount of Ingredients**
1 ¼ cup of powdered milk	
¼ cup condensed milk	
¼ cup sugar	
Oil, for molding	

Sample Only: I need to make ***10*** pastillas.

(I use 10 so that it is not a multiple of 3)

Ratio: the amount of ingredient ***to*** the number of pastillas it produces

$$1\frac{1}{4} : 30 = n : 10 \quad \Rightarrow \quad \frac{5}{4} : 30 = n : 10 \quad \Rightarrow \quad 30n = \frac{50}{4}$$

$$n = \frac{50}{4} \div \frac{30}{1} = \frac{50}{120} = \frac{5}{12}$$

Therefore, I need $\frac{5}{12}$ cup of powdered milk.

MODULE 4

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Business Mathematics. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of learners. The lessons are arranged to follow the standard sequence of the course.

After going through this module, you are expected to:

1. recognize the type of proportion in the problem.
2. use fundamental property of proportions to solve a proportion; and
3. solve problems involving direct, inverse, and partitive proportions.

Lesson 4 Solving Problems Involving Proportions

Proportion is not just a mathematical term confined in a book or in a classroom setting. You apply it every day without even realizing that you are actually dealing with it. In this module, you will use the concepts you have learned in the previous module. You will solve different problems that will allow you to see proportions as part of your daily life.

What is It

To solve any proportion problems, it would be easier if you write the given in the form of a mathematical statement. Since it is a proportion, remember to set up the two ratios in order. Meaning, each term of the first ratio corresponds to each term of the second ratio.

A. Direct Proportion

Example 1: If you work 15 hours, you earn Php 600. How much will you earn if you work 60 hours?

- The first set of ratio is 15 hours of work to Php 600.
- The second ratio is 60 hours of work to ***n*** pesos.

You may set the proportion in any or the vice versa of the following ways:

$$\frac{hours\ of\ work_1}{amount\ of\ earning_1} = \frac{hours\ of\ work_2}{amount\ of\ earning_2}$$

$$\frac{hours\ of\ work_1}{hours\ of\ work_2} = \frac{amount\ of\ earning_1}{amount\ of\ earning_2}$$

$$\frac{amount\ of\ earning_1}{amount\ of\ earning_2} = \frac{hours\ of\ work_1}{amount\ of\ earning_2}$$

Steps	Solution
1. Identify the type of proportion.	It is a direct proportion because ***more*** hours of work means ***more*** earnings.
2. Translate to a proportion statement.	$\frac{15}{600} = \frac{60}{x}$
3. For a direct proportion, simply use the cross-product method.	15x = 36000 $\frac{15x}{15} = \frac{36000}{15}$ x = 2400
4. Write a concluding statement.	You will earn Php 2400 for 60 hours of work.

Example 2: Three box of candies cost Php 195. How much do 8 boxes of candies cost?

Steps	Solution
1. Identify the type of proportion.	It is a direct proportion because ***more*** boxes of candies means ***higher*** cost.
2. Translate to a proportion statement.	$\frac{3}{195} = \frac{8}{x}$
3. For a direct proportion, simply use the cross-product method.	3x = 1560 $\frac{3x}{3} = \frac{1500}{3}$ x = 520
4. Write a concluding statement.	Eight boxes cost Php 520.00.

B. Inverse Proportion

Example 1: A supply of food is enough to feed 10 persons for 2 days. If 30 more people join the camp, how long will the food supply last?

If you solve this using the method you used in direct proportion,

$\frac{10\ persons}{2\ days} = \frac{40\ persons}{x}$ or $\frac{10}{2} = \frac{40}{x}$ ➡ 10x = 80 ➡ x = 8

What?! How come that when there are 10 people, the supply is enough for 2 days, but if there are 40 people, this supply is enough for 8 days?

Let us analyze the problem. Let us assume that the food supply is 10-kg rice. What do you think will happen with this supply if 30 more people will be added to 10 people? Will it still be enough in two days? We can see in this problem that as the number of people ***increases,*** the number of consumption days ***decreases***. Therefore, this is an inverse proportion.

So, if the method you use in direct proportion will not work in this type of proportion, what will you do to solve this?

Steps	Solution
1. Identify the type of proportion.	It is an inverse proportion because as the number of people ***increases***, the number of consumption days ***decreases***.
2. Translate to a proportion statement.	Since it is an inverse proportion, we will not use the cross-product method. Instead, we will use $x_1y_1 = x_2y_2$ (10)(2) = (40)(n)
3. Solve for n.	10(2) = 40n or 40n = 20 $\frac{40n}{40} = \frac{20}{40}$ n = $\frac{1}{2}$
4. Write a concluding statement.	If there are 40 persons, it will only take half day to consume the food supply.

Example 2: If 3 men can paint a room in 8 days, then how many men are needed to finish the same job in 4 days?

Steps	Solution
1. Identify the type of proportion.	It is an inverse proportion because as the number of people ***increases***, the number of working days ***decreases***.
2. Translate to a proportion statement.	We will use $x_1y_1 = x_2y_2$ (3)(8) = (4)(n)

3. Solve for n.	3(8) = 4n or 4n = 24 $\frac{4n}{4} = \frac{24}{4}$ n = 6
4. Write a concluding statement.	6 men are needed to finish the job in 4 days.

C. Partitive Proportion

Example 1: Forty (40) students took a mathematics test after finishing a course. How many failed in this test if the ratio of those who passed to those who failed the test was 6:2?

Steps	Solution
1. Identify the type of proportion.	This is a partitive proportion because the whole 40 students were divided into parts – those who passed and those who failed.
2. Translate to a mathematical statement.	Let n be the GCF. 6n + 2n = 40
3. Since the GCF is unknown, find its value by solving for the unknown.	those who fail those who pass Note that 6n + 2n is equal to 40 students. 6n + 2n = 40 8n = 40 $\frac{8n}{8} = \frac{40}{8}$ n = 5
4. Remember that you are only asked for those who "failed" in the test. Thus, you need to know 2n.	2n = 2 (5) = 10
5. Make a statement.	There were 10 students who failed in the test.

Example 2: Agents X, Y, and Z were able to sell 225 boxes of candies in the ratio 2:3:4 respectively. How many boxes of candies did each agent sell?

Steps	Solution
1. Identify the type of proportion.	This is a partitive proportion because the whole 225 boxes are divided into three parts in the ratio 2:3:4.

2. Translate to a mathematical statement.	Let n be the GCF. 2n + 3n + 4n = 225
3. Since the GCF is unknown, find its value by solving for the unknown.	9n = 225 $\frac{9n}{9} = \frac{225}{9}$ n = 25
5. Remember that you are asked for the number of boxes sold by each agent.	Agent X: 2n = 2(25) = 50 Agent Y: 3n = 3(25) = 75 Agent Z: 4n = 4(25) = 100 Total = 225
6. Make a statement.	Agent X sold 50 boxes of candies. Agent Y sold 75 boxes of candies. Agent Z sold 100 boxes of candies.

What's More

Activity: Words to Symbols

Directions: Translate each of the following problems into a mathematical statement. Use n for the unknown.

1) A piece of cloth, 500 cm long, is divided into two pieces in the ratio 4:3. How long is the shorter part?

2) Jenny earns Php 500 a day. How much does she earn if she works 15 days?

3) 10 workers are needed to finish a certain job in 24 hours. How long will it take 15 workers to finish the same job?

4) An encoder can finish 4 pages in 60 minutes. How long will it take him finish 20 pages?

5) If 20 people can do a job in 8 days, how many days will 5 people do the same job?

6) At the supermarket, oranges are being sold for Php 72 for every 8 pieces. How much will you pay if you buy 24 oranges?

What I Can Do

Instructions:

1. Your parents gave you a 100-peso bill for your food and cellphone load allowance. How much will you spend on each item?

Item	Amount
Food	
Cellphone Load	
Total	***Php 100.00***

2. Based on your budget, make a partitive proportion statement inside the box below. Remember to express the ratio in simplest form.

   Example:
   I will divide my Php 100 for my food and cellphone load in the ratio 3:5.

Rubric:

	3 points	2 points	1 point	SCORE
Mechanics	All instructions were followed.	Has committed 1 error in following the instructions	Did not follow any of the instructions	
Content	Shows clear understanding of the mathematical concept	Shows some understanding of the mathematical concept	Shows no understanding of the mathematical concept	
Organization	The statement is clear and easy to understand.	The statement is somehow clear and easy to understand.	The statement is difficult to understand.	
Total				

MODULE 5

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Business Mathematics. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of learners. The lessons are arranged to follow the standard sequence of the course.

After going through this module, you are expected to:

1. differentiate markup, markdown, and mark on; and
2. obtain markup, markdown, and mark on given the price of a product.

Lesson 5 Markup, Markdown, and Mark On

Pricing plays a vital role in increasing your business' profits. Hence, a poor pricing strategy leads to hurting your sales drastically. This lesson will help you learn the different terms needed in setting the price.

What is It

Some people think that the terms "cost price" and "selling price" are synonyms. However, they are not the same in context. **Cost price** is the amount a business spends to produce goods or services before adding a profit. It includes the price of the raw materials used in creating a product. Usually, it is on a per-unit basis. **Operating cost** is the expense relative to the production and actual selling of a product. It comprises the salaries, rental fees, transportation, and other incidental payments; it does not include the amount incurred for the raw materials. These two terms (cost price and operating cost) are necessary for determining the **selling price** – the amount a customer is willing to pay for the product or service offered by the business. In determining the price, a business owner must set a target profit after the sale. Thus, the formula to compute the selling price is:

$$\boxed{S = C + E + P}$$

where: S = Selling Price
C = Cost Price
E = Operating Cost/Expense
P = Profit

The amount added to the cost price is called **markup**. It is the difference between the selling price and the cost price, and it is computed based on the cost. To determine the markup, use the formula:

MU = S – C
or
MU = E + P

where:
MU = Markup
S = Selling Price
C = Cost Price
E = Operating Cost/Expense
P = Profit

Example: Jenny would like to sell pad papers she purchased from a mall. The cost of pad paper is Php 12.00 each. The operating cost is set at 15% of the cost and she would like to have a 25% profit on the cost of each item.

a. Determine the markup price for each pad paper.
b. Help Jenny compute the selling price for each pad paper.

Solution:
a. Determine the markup price for each pad paper.

Step	Solution
Take note that markup is the sum of the operating expense and profit.	MU = E + P MU = 15% of C + 25% of C MU = 0.15(12) + 0.25(12) MU = 1.8 + 3 MU = Php 4.80
Therefore, the markup for each pad paper is Php 4.80.	

b. Help Jenny compute the selling price for each pad paper.

Step	Solution
The formula S = C + E + P can also be transformed to S = C + MU.	S = C + MU S = 12 + 4.80 S = Php 16.80
Therefore, the selling price for each pad paper is Php 16.80.	

Now, some products increase their price during peak seasons because the demand for them rises. For instance, the price for red roses spikes during February 14. The increase on top of the selling price when there is seasonal demand for the product or an undersupply of the product due to calamity is known as the mark on.

Example: The cost per unit of a flower is Php 50.00. The owner added a 45% markup for sale. However, a mark on of 25% of the regular selling price is added on it during peak seasons.

a. How much is the regular selling price (S)?
b. How much is the selling price during peak seasons (PS)?

Solution:
a. How much is the regular selling price (S)?

Steps	Solution
1. Take note that markup is computed based on the cost price.	MU = 45% of the cost price MU = .45 (50) MU = Php 22.50
2. Compute the selling price using the available data.	S = C + MU S = 50 + 22.50 S = Php 72.50
Therefore, the regular selling price of a flower is Php 72.50.	

b. How much is the selling price during peak seasons (PS)?

Steps	Solution
1. Compute the mark on. The problem states that the mark on is based on the regular selling price.	MO = 25% of the regular selling price MO = .25 (72.50) MO = Php 18.15
2. Find the selling price during peak seasons.	PS = S + MO PS = 72.50 + 18.15 PS = Php 90.65
This means that each flower sells for Php 90.65 during peak seasons.	

In some cases, businesses cut down prices to generate more sales or clear inventory. It happens when they sell the product at a discounted price or sale price. This decrease in the selling price is called markdown. Unlike markup, markdown is computed based on the selling price. To calculate, subtract the discounted price from the regular selling price. In symbols,

$$\boxed{\mathbf{MD = S - SP}}$$

where: MD = Markdown
S = Selling Price
SP = Sale Price/Discounted Price

Example: Enzo had the chance to buy a branded shirt regularly prices at Php 990.00 for only Php 693.00.

a. What was the amount of the markdown?
b. What was the rate of the markdown?

Solution:
a. What was the amount of the markdown?

Steps	Solution
Remember that markdown is the difference between the regular selling price and the discounted/sale price.	MD = S – SP MD = 990 – 693 MD = Php 297.00
This means that markdown price is Php 297.00	

b. What was the rate of the markdown?

Steps	Solution
Take note that markdown is computed based on the selling price.	$MDr = \frac{MD}{S} * 100$ $MDr = \frac{297}{990} * 100$ MDr = 0.3 * 100 MDr = 30%
This means that markdown rate is 30%.	

What's More

Read the following situation and supply the needed information.

Rose bought boxes of chocolates for Php 85.00 each. She decided to impose a markup rate of 40% on each box before selling it to the market. During holidays, a mark-on of 15% of the regular selling price is added on. One day during a regular season, she found out that the chocolates are about to expire. To dispose of the products immediately, she decided to lower the price by Php 29.75.

1. What is the cost price of each box of chocolate?
2. Compute the markup.
3. How much is the price per unit during holidays?
4. How much is the sale price?
5. What is the markdown rate?

What I Can Do

Activity: Find it Out!

1. A store owner used a markup rate of 50%. How much is the markup if the item costs the owner Php 300.00?

Solution	**Final Answer**

2. During a sale, a shirt regularly priced at Php 500.00 was sold at a 15% markdown. How much was the markdown?

Solution	**Final Answer**

3. Many people prefer to eat fish over pork and beef during a special season. Because of the high demand, fish vendors increase the price by 35% of the selling price. How much is the new selling price if the regular selling price is Php 110.00 per kilo?

Solution	**Final Answer**

MODULE 6

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Business Mathematics. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of learners. The lessons are arranged to follow the standard sequence of the course.

The module is divided into two lessons, namely:

- Lesson 6.1 – Markup and Margin
- Lesson 6.2 – Trade Discounts

After going through this module, you are expected to:

1. differentiate mark-up from margins.
2. describe how gross margins is used in sales; and
3. compute single trade discounts and discount series.

Lesson 6.1 Markup and Margin

The aim of any business is to succeed in its chosen field. However, success in business does not only mean having a high revenue or gross sales. It is also about how much is the profit it gained from selling its products or services. In this lesson, you will know some ways how businesses compute the rate of **gross profit** – the amount left when you subtract the cost of goods sold from the revenue or sales. **Cost of goods sold (COGS)** is the direct expenses incurred in manufacturing a product or delivering a service whereas the **revenue** (or sales) is the income a company earns solely from selling the products or services.

What is It

Markup and margin have commonalities. Both margin and markup tell you the difference between the selling price and COGS, but they differ on perspectives. Recall that markup is the amount you added to the cost price before selling it to the consumers. It tells you about the relationship between your selling price and the amount it costs you to manufacture a product or provide a service. On the other hand, margin (or gross margin) is the amount you gain on making a sale or service. It shows you the relationship between sales and profits. Hence, it reveals more precise data about your actual earnings. Thus, the higher the markup, the higher the selling price. The higher the margin, the higher the profits.

However, to calculate the **rates** of markup and margin:

Markup	**Margin**
$\text{MU}(\%) = \frac{\text{Gross profit}}{\text{Cost of Goods Sold}} \text{ x } 100$	$\text{M}_s(\%) = \frac{\text{Gross profit}}{\text{Revenue}} \text{ x } 100$

Example: MXY Company sells laptops worth ₱31,999 each. Each unit costs 85% of its selling price. What are the gross profit of the product, the rates of markup rate and margin?

Solution:

Steps	**Solution**
Identify the given.	S = ₱31,999.00 C = 85% of S C = 0.85 (31,999) **C = ₱27,199.15**
Find the gross profit.	GP = 31,999 – 27,199.15 **GP = ₱4,799.85**
Calculate the rate of markup.	$\text{MU}(\%) = \frac{\text{Gross profit}}{\text{Cost of Goods Sold}} \text{ x } 100$ $\text{MU}(\%) = \frac{4799.85}{27199.15} \text{ x } 100$ $\mathbf{MU}(\%) = \mathbf{17.65}\%$
Calculate the rate of margin.	$\text{M}_s(\%) = \frac{\text{Gross profit}}{\text{Revenue}} \text{ x } 100$ $\text{M}_s(\%) = \frac{4799.85}{31999} \text{ x } 100$ $\mathbf{M_s}(\%) = \mathbf{15}\%$

What's More

Activity: Complete the Table

Revenue	COGS	Gross Profit	Markup (R)	Margin (R)
A. ₱3,500	₱1,200			
B. ₱4,275	₱2,327			
C. ₱7,000	₱6,000			
D. ₱10,500	₱4,850			

Lesson 6.2 Trade Discounts

Suppliers make different strategies to attract people to patronize their products or services. One of their tactics is to offer discounts to customers. It is to encourage more resellers and to keep their customers loyal to them. In this lesson, you will learn how to compute trade discounts and discount series.

What is It

Suppliers always set a suggested retail price for their products. It is called the list price. However, they usually give a special discount to those people involved in bringing these to the customers. Those people are called the middlemen, and the discount they receive is known as a trade discount. This price reduction will allow the middlemen to gain a profit in selling the products.

To compute for the trade discount:

Formula	Description
$TD = TD_R\ (S)$	TD = Trade Discount TD_R = Trade Discount Rate S = Selling Price

Example: Your regular pork supplier gives you a 10% discount on your weekly purchases. How much is the trade discount if the price of pork per kilo is ₱380.00 and you are planning to buy 7 kilograms next week?

Solution:

Step	Solution
1. Compute the trade discount per unit.	TD = TD_R (S) TD = .10(380) TD = ₱38.00 per kilo
2. Multiply the computed trade discount per kilo by 7.	₱38.00 (7) = ₱266.00
Therefore, you will receive a total trade discount of ₱266.00.	

There are times that a supplier gives a series of discounts to activate a specific behavior from the customers. For example, the supplier's desired behaviors from the clients are these:

- Clients will choose to buy from them regardless of the competition.
- Clients will buy items in volume.
- Clients will pay in cash.

In this case, the supplier may give a discount of 15/15/10. The symbol you see represents a trade discount series.

For instance, a facemask supplier wants to give its distributors a discount of 20% when they buy during morning hours. In addition to the 20% discount, the distributors will receive a 10% discount if they purchase a set minimum volume of items and another additional 10% if they pay in cash. In symbol, 20/10/10.

Suppose that the required minimum order is 1000 boxes of the facemask. How much will you pay in cash if you buy 1,200 boxes that cost ₱35.00 each? Assume that you will do the transaction during morning hours.

Solution:

<table>
<tr><th>Traditional Method</th><th>Single Equivalent Rate (SER) Method</th></tr>
<tr><td>1200 boxes (₱35) = ₱42,000
Note that there are three offered discounts here.

Step 1: Find the first discount (D_1).
₱42,000 (.20) = ₱8,400
Hence, the new price to pay is
₱42,000 - ₱8,400 = ₱33,600

Step 2: Find the second discount (D_2).
₱33,600 (.10) = ₱3,360
So, the new price to pay is ₱30,240.

Step 3: Find the third discount (D_3).
₱30,240 (.10) = ₱3,024

Finally, the net price to pay is ₱27,216.</td><td>In this method, you must know the discount complement. It means you need to subtract the given discount rate from 100%.

The given trade discount series in the problem is 20/10/10. Find the rate of discount complement.

<table>
<tr><th>Discount (D_r)</th><th>Discount Complement (C_r)
100% - D_r</th></tr>
<tr><td>20%</td><td>80%</td></tr>
<tr><td>10%</td><td>90%</td></tr>
<tr><td>10%</td><td>90%</td></tr>
</table>

SER = 1 – [(C_{r1})(C_{r2})]
SER = 1 – [(.80)(.90)(.90)]
SER = 1 – .648</td></tr>
</table>

	SER = 0.352 or 35.2% Meaning, a discount series 20/10/10 is NOT the same with 40%. The single equivalent rate of 20/10/10 is 35.2%. Total Discount = 42,000 (.352) Total Discount = ₱14,784.00 Hence, the net price is ₱42,000.00 - ₱14,784.00 = ₱27, 216.00

What's More

Activity: Complete the table.

List Price	Discount Series	Complement	SER	Total Discount	Net Price
₱1,750.00	10/7/5				
₱3,100.99	20/12				
₱8,499.00	15/10/5				
₱10,575.00	12/10/7				

MODULE 7

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Business Mathematics. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of learners. The lessons are arranged to follow the standard sequence of the course.

After going through this module, you are expected to:

1. differentiate profit from loss.
2. illustrate how profit is obtained and how to avoid loss in a given transaction.
3. determine the break-even point; and
4. solve problems involving buying and selling products.

Lesson 7 Profit and Loss

In the previous module, you have learned that people in the business sector offer discounts to attract and keep more customers. They do this to make more sales and increase their profit. But how to know if the company is making a profit? In this lesson, you will learn how to determine whether a business is having a profit or not.

What is It

Profit is the amount left when you subtract all the expenses from the revenue (or sales). We have different types of profit:

Sales – Cost of Goods Sold = Gross Profit
Gross Profit – Expenses (Operational) = Net Profit

There are instances that when you subtract the expenses from the sale, the result is a negative number. It occurs when the expenses exceed the revenue (or sales). In this case, it is a **loss** for the company. In other words, there is a profit if the difference between the sales and expenses is positive. If it is negative, then it is a loss.

A company may also experience a break-even point. It happens when the profit is zero. In this case, the revenue is equal to the expenses. The break-even point helps to determine the minimum number of items needed to sell to avoid losses. The sales made with the break-even point are called the break-even sales. To calculate this, multiply the selling price by the break-even point.

Example: Phoebe sells dolls. The labor, material, packaging, and shipping cost her ₱500 per doll. Suppose that the total expenses incurred to manufacture the dolls are ₱1,250,000, and the dolls are sold for ₱1,000 each.
a. How much is the revenue if 2,500 pieces of dolls are sold?
b. If she was able to sell only 1,200 pieces of the doll, was it a loss or a profit?
c. What sales volume is necessary for Phoebe to break even?

Solution:

a. How much is the revenue if 2,500 pieces of dolls are sold?

Step	Solution
1. Identify the given.	Quantity = 2,500 pieces Selling Price = ₱1,000
2. Compute the revenue.	R = ₱1,000 (2,500) R = ₱2,500,000

b. If she was able to sell only 1,200 pieces of doll, was it a loss or a profit?

Step	Solution
1. Compute the revenue for 1,200 pieces of dolls.	R = ₱1,000 (1,200) R = ₱1,200,000
2. Subtract the expenses from the revenue.	R – E = Profit/Loss ₱1,200,000 - ₱1,250,000 = -₱50,000
Since the result is a negative number, then it is a loss. It means that Phoebe will lose ₱50,000 if she will only be able to sell 1,200 pieces of dolls.	

c. What sales volume is necessary for Phoebe to break even?

Step	Solution
Compute the break-even point. Note that to compute the revenue, we multiply the quantity by the selling price. Since we do not know yet the required quantity, we will use a variable N.	Revenue = Expenses ₱1,000N = ₱1,250,000 $\frac{1000N}{1000} = \frac{1,250,000}{1000}$ N = 1,250
Therefore, to make a break-even sale, Phoebe should be able to sell 1,250 pieces of dolls.	

What's More

Activity: Profit or Loss?

Directions: Find out if the business made a profit or loss.

Revenue	Expenses	Amount of Profit/Loss	Remark (Profit or Loss)
1) ₱2,800.00	₱2,499.00		
2) ₱4,500.00	₱4,650.00		
3) ₱10,875.00	₱10,892.00		
4) ₱15,186.00	₱12,123.00		
5) ₱24,432.00	₱26,900.00		

What I Can Do

Activity: Profit or Loss?

Directions: Find the break-even point (the minimum number of items needed to be sold) to avoid losses.

Given	Break-even Point
1. The expenses incurred in manufacturing a bag is ₱270,000.00 while the selling price of a bag is ₱300.00.	
2. Each ballpen is being sold at ₱ 20.00. The amount spent in producing the ballpens is ₱1,260.00.	
3. A company spent ₱100,000.00 in manufacturing its beauty products. The selling price of each product is ₱250.00.	

MODULE 8

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Business Mathematics. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of learners. The lessons are arranged to follow the standard sequence of the course.

After going through this module, you are expected to:

- compute interest specifically as applied to mortgage, amortization, and on services/utilities and on deposits and loans

Lesson 8 Mortgage and Amortization

If you want to buy a property but you do not have enough money to pay it off in full, banks or other lending institutions offer different types of loans to help you. In return, they will charge you an amount, called the **interest**, for using their money. In this lesson, you will learn more about how it works in mortgage.

What is It

Mortgage (*pronounced with silent T*) is a type of loan you use to buy real estate properties. For instance, if you want to buy a house and lot but your savings is not enough to pay the property in full, you may apply a mortgage from a bank or any lending institutions. Of course, these institutions will secure a collateral from you. **Collateral** is a property that the lenders will take from you just in case you cannot pay them anymore for some reasons. Oftentimes, the collateral is the property you acquired using the funds from the mortgage. Most of the time, the lender will also require you to pay for a **down payment**. Note that the higher the down payment, the better. Why? Because it will decrease the amount of mortgage you will avail from the lender. Note that these lenders will charge you an interest only for the amount you loan from them. Let us say that you give a 40% down payment. It means that the remaining amount or the **mortgage loan** will just be 60% of the total contract price (TCP). Hence, only 60% of the TCP will be subject for interest.

Example 1: You are planning to buy a house and lot from a real estate company worth ₱2,000,000. However, your savings is not enough to fully paid the property, so you applied for a mortgage from a bank. The bank required you to make a 20% down payment.

a. How much is the down payment?
b. How much is the mortgage loan?

Step	Solution
1. Identify the given	TCP = ₱2,000,000 Down payment = 20% of the TCP
2. Calculate the amount of down payment.	Down payment = TCP (down payment %) Down payment = 2,000,000 (.20) Down payment = ₱ 400,000.00
3. Compute the mortgage loan.	Mortgage Loan = TCP – Down Payment Mortgage Loan = 2,000,000 – 400,000 Mortgage Loan = ₱1,600,000.00
Therefore, the required down payment is ₱400,000.00 and the mortgage loan is ₱1,600,000.00.	

In the example above, the amount left after you give your down payment is ₱1,600,000.00. It means that you will be paying this amount to the lender on an installment basis, usually a series of fixed payments, called **amortization**. In this case, the lender will spread out the repayment of loan over an agreed period of time referred to as the **term** of the loan. Part of the amortization you pay goes to the principal, and a portion of it goes to the interest. **Principal** refers to the original amount you borrowed from the lender. To determine the monthly amortization, use the formula:

$$A = \frac{iP(1+i)^n}{(1+i)^n - 1}$$

where: A = monthly amortization
P = principal
i = monthly interest rate
n = total number of payments

Example 2: Suppose that you will pay the mortgage loan for 25 years. How many months will it take you to fully pay your loan?
Solution:

Term of the Loan = 25 years
25 years (12 months per year) = 300 monthly payments
Therefore, the total number of payments to fully pay the loan is 300 months.

Example 3: Using the previous example, the mortgage loan of ₱1,600,000 is subject for a 7.5% annual interest rate and you need to pay this for 25 years.

a. How much is the monthly payment?
b. How much is the interest incurred for 25 years?

Step	Solution
1. Identify the given.	P = ₱1,600,000 n = 300 annual interest rate = 7.5% or 0.075 i = .075/12 = .00625
2. Solve using the formula for monthly amortization.	$A = \frac{iP(1+i)^n}{(1+i)^n-1}$ $A = \frac{(.00625)(1{,}600{,}000)(1+.00625)^{300}}{(1+.00625)^{300}-1}$ $A = \frac{(10{,}000)(1.00625)^{300}}{(1.00625)^{300}-1}$ $A = \frac{(10{,}000)(1.00625)^{300}}{(1.00625)^{300}-1}$ $A = \frac{(10{,}000)(6.4829)}{6.4829-1}$ $A = \frac{64829}{5.4829} = 11{,}823.85$
Thus, you will pay ₱11,823.85 monthly for 25 years.	
3. Calculate the total amount of payment you made for 25 years.	Total Payment = A (n) Total Payment = 11,823.85 (300) Total Payment = ₱3,547,155.00
4. Subtract the mortgage loan from the total amount of payment.	Total Interest = Total Payment – Mortgage Loan

	Total Interest = ₱3,547,155.00 - ₱1,600,000.00 Total Interest = ₱1,947,155.00
It means that the total interest incurred for 25 years is ₱1,947,155.00.	

What's More

Activity: Find It Out!

Directions: You bought a house and lot from a real estate company in the Philippines. The total contract price of your chosen property is ₱2,000,000, and you are required to pay a 20% spot down payment. Suppose you applied for a mortgage from a bank that offers a 7.5% annual interest rate to pay off the remaining amount, and you are planning to pay the loan for 20 years. Find out the following:

1. Amount of down payment : _____
2. Mortgage Loan Amount : _____
3. Number of payments : _____
4. Monthly amortization : _____
5. Interest incurred for 20 years : _____
6. Total Payment for 20 years : _____

What I Can Do

Activity: Choose Wisely!

Scenario: The total contract price of a house and lot is ₱2,500,000.00. You paid a 20% spot down payment, and to pay the remaining 80%, you applied for a mortgage from a lending institution that offers an 8.1% annual interest rate. Compute the monthly amortization and the total interest accumulated based on the term indicated in the first column.

Term	Monthly Amortization	Accumulated Interest
10 years		
15 years		
20 years		
25 years		
30 years		

Question: Do you think it is wise to extend the term to a longer term if you have the capacity to pay the loan in a short span of time? Why or why not?

_____
_____
_____
_____

MODULE 9

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Business Mathematics. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of learners. The lessons are arranged to follow the standard sequence of the course.

This module targets the following learning competencies:

1. Illustrate the different types of commissions (ABM_BM11BS-IIa-11).
2. Compute commissions on cash basis and commissions on installment basis (ABM_BM11BS-IIa-12).
3. Compute down payment, gross balance and current increased balance (ABM_BM11BS-IIa-13).

After going through this module, you are expected to:

- define commission;
- define each of the types of commission;
- compute commissions on cash basis;
- compute commissions on installment basis; and
- define and compute down payment, gross balance, and current increased balance.

Lesson 9 Commissions

There are many reasons why people work. One of these is to make money. Some earn this by establishing their own business, while some by receiving compensations for their service through a salary, a wage, or an incentive and/or commission. The latter will be the topic of this module. You will find out the different types of commissions and how it is being computed.

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What is It

Compensations for a person's service are not always fixed. Sometimes, people get paid based on the percentage of selling price of the products or goods they sell. The fee that these people are receiving is called a commission. This serves as a payment in exchange for a person's service in either managing or completing a sale. Examples of commission-based businesses are real-estate marketing projects, car dealership, and direct selling companies. Whenever a salesperson closed a deal, he or she will get a commission from it.

Example: A direct seller of skin care products was able to sell 25 bottles of whitening lotions amounting to Php 75.00 per piece. How much was she earning if the company gave her a 25% commission rate for the products she sold?

Solution:

Each bottle of lotion costs Php 75.00. She was able to sell 25 bottles of it.

Steps	Solutions
Find the total amount of sold products.	25(Php 75) = Php1875
Express the rate to decimal form. Multiply the total amount by the commission rate.	Express 25% to decimal: 0.25 Php 1875 (0.25) =Php 468.75
State the final answer.	The direct seller received Php 468.75 as commission for her service to the company.

Types of Commission

Generally, there are three types of commission.

1. **Straight Commission**

   This type of commission is also known as the revenue commission because the fee is only based on the percentage of sales. Meaning, the employee receives no fixed salary because everything depends on her performance to sell products. It is advantageous if the person is selling high-cost items.

   Example:

Jen receives a 25% commission on the beauty products she sells. If she sells a lotion for Php 350, 2 whitening soaps for Php 65 each, and a set of makeup for Php 975, how much does she make in commission?

Solution:

Steps	Solutions
Find the total amount of sold products.	=Php 350 + 2(Php 65) + Php 975 =Php 1455
Express the rate to decimal form. Multiply the total amount by the commission rate.	Express 25% to decimal: 0.25 Php 1455 (0.25) =Php 363.75
State the final answer.	Jen will receive Php 363.75 as commission for her service to the company.

2. Salary Plus Commission

Some company offers this type of commission wherein the salesperson receives a basic salary plus a percentage of whatever sales he makes. In other words, the employee gets paid (salary) even without making any sales.

Example:

Enzo works in a company that pays Php 2,500 a week and a 12% commission on sales above Php 15,000 for the week. How much will he receive if he sold products worth Php 28,000 in a week?

Solution: Take note that the problem states that he will only receive a commission based on his sales above Php 15,000.

Steps	Solutions
Subtract Php 15,000 from the amount of sold products.	=Php 28,000 – Php 15,000 =Php 13,000
Multiply the result by the commission rate.	Express 12% to decimal: 0.12 Php 13,000 (0.12) = Php 1,560
Add the commission to the salary.	Php 2,500 + Php 1,560 = Php 4,060
State the final answer.	Enzo will receive Php 4,060.00 for that week.

3. Graduated Commission

This type of commission has a step-up fee structure because the commission varies depending on how much sales are made. Whenever a salesperson achieves a sales milestone, the commission rate gets higher.

Example:

A company offers the following commission structure:

5% on the first Php 15,000 sold

10% on the next Php 20,000 sold

15% on all sales beyond Php 35,000

How much is the gross pay if the salesperson sells Php 75,000?

Steps	Solutions
Compute the first commission. Remember to express the rate to decimal form.	Php 15,000 (.05) = Php 750
Compute the second commission.	Php 20,000 (.1) = Php 2,000
Compute the third commission.	All sales beyond Php 35,000 mean all sales above Php 35,000. Php 75,000-35,000 = Php 45,000 Php 45,000 (.15) = Php 6,750
Add all the commissions.	Php 750 + Php 2,000 + Php 6,750 =Php 9,500.00
State the final answer.	The salesperson will receive a gross pay Php 9,500.

B. Computing Commissions on Cash Basis and Commissions on Installment Basis

1. Commission on Cash Basis

The computation is the same as the computation of straight commissions.

Example:
Jen is a salesperson at XYZ Computer Store. She gets a 6.1% commission for every cash purchase of a laptop. In her first month of work, she was able to sell five laptops that were paid cash at Php 36,000 each. How much was her total commission for such sales?

Solution:

Steps	Solutions
Find the total amount of sold products.	= 5 (Php 36,000) = Php 180,000
Multiply the commission rate by the total amount of sold products.	6.1% is 0.061 in decimal form. = Php 180,000 (0.061) = Php 10,980
State the final answer.	Jen received a Php 10,980 commission for all the cash purchase of laptop.

2. Commission on Installment Basis

In some cases, customers opt to pay for the purchased item on an installment basis.

Example:

Dolfo is a salesperson at XYZ Computer Store. He was able to sell five laptops that cost Php 36,000 each. The promo is that the customer may pay the amount in six (6) months wherein the total price is equally divided into six equal monthly installments with 0% interest. Dolfo gets a 2% commission on the first month of payment, and it decreases by 0.30% every month thereafter. The commission is computed based on the outstanding balance for the month. How much will be his total commission at the end of the installment period?

Solution:

Since the commission varies every month for six months, we must find out the commission he received for each month.

Php 36,000 ÷ 6 = Php 6,000 monthly fee

- First month commission
  = Php 36,000/laptop x 5 laptops x 0.02 = Php 3,600
- Second month commission
  = Php 30,000/laptop x 5 laptops x (0.02 – 0.003) = Php 2,550
- Third month commission
  = Php 24,000/laptop x 5 laptops x (0.017 – 0.003) = Php 1,680
- Fourth month commission
  = Php 18,000/laptop x 5 laptops x (0.014-0.003) = Php 990
- Fifth month commission
  = Php 12,000/laptop x 5 laptops x (0.011 – 0.003) = Php 480
- Sixth month commission
  = Php 6,000/laptop x 5 laptops x (0.008 – 0.003) = Php 150

Dolfo's total commission for six months is Php 9,450.00

C. Down Payment

A down payment is an initial payment you give when you buy something pricey or expensive. This payment is deducted from the purchase price. You minimize the remaining balance to the company when you provide a down payment. The higher the down payment, the lower the remaining balance is. For example, when you buy a house in the future and you make a 20% down payment, then all you need to loan is 80% of the total price. When you give a 60% down payment, then your remaining balance is just 40% of the purchase price. Most of the time, this remaining balance is subject to interest.

Example:

A customer wants to buy a car amounting to Php 1,100,000. The agreement with the company is that the customer should submit all the requirements with a down payment before taking home the car. How much are the down payment and the remaining balance if the customer will give a 30% down payment?

Solution:

Steps	Solutions
Compute the down payment first.	The customer gives a 30% down payment. Just multiply the price by 30% or 0.3. = Php 1,100,000 (0.3) = Php 330,000
Subtract the down payment from the total price of the car.	= Php 1,100,000 – Php 330,000 = Php 770,000
Write a concluding statement.	The customer will pay Php 330,000 as down payment. Thus the remaining balance is Php 770,000.

D. Book Balance or Gross Balance

Book balance is a banking term that refers to the actual money accessible for a person or a company to use. It is "a measure of what the bank has on hand before adding or subtracting regulatory obligations and items that will soon appear on its books." (Paredes et al., 2016)

Example:

Your bank passbook currently has Php 50,000. This day, a client deposited a check amounting to Php 15,000 in your bank account. Because of this, the reflected amount in your bank passbook becomes Php 65,000. However, you cannot withdraw the whole amount of Php 65,000 today because the deposited check has not cleared yet. Your actual money in the bank (which is the gross balance) is only Php 50,000 because the check has not been cleared yet.

E. Current Increased Balance

Some clients fail to pay loans on time. This is now where the term "current increased balance" comes in. This term is used to describe the total amount a client needs to pay that includes the interests or penalties earned by unpaid loan balance that is supposed to be paid but were not able to make it on time.

Example:

The total current amount due from a purchase made using a credit card is Php 55,286.50. Based on the agreement, the minimum required payment is 7.5% of the total amount due. If the client pays only the minimum required payment, a charge of 3% of the remaining balance will be carried on to the next bill. If there are no purchases made for the next three months and a charge of 3% is added every billing period, what is the required monthly bill for the next three months?

Solution:

Time (month)	Total Amount Due for the Month	Minimum Required Payment for the Month
0	55,286.50	0.075 (55,286.50) = 4,146.49
1	= (55,286.50 – 4,146.49) + 0.03 (55,286.50 – 4,146.49) = 51,140.01 + .03(51,140.01) = 1.03(51,140.01) = 52,674.21	0.075 (52,674.21) = 3,950.57
2	=1.03(52,674.01 – 3950.57) =1.03(48,723.61) =50,185.35	0.075 (50,185.35) = 3,763.90
3	=1.03(50,185.35 – 3763.90) =1.03(46,421.45) =47,814.09	0.075 (47,814.09) =3,586.06

Based on the table, you will see that the client paid Php 11,300.53 for the next three months. However, if you will look at the second column, only Php 7,472.41 has been deducted from the original debt of Php 55,286.50. It shows that more payment goes to interest fees rather than to the sum of the actual debt every time a client is paying only the required minimum payment. The highlighted amount in the second column is the current increased balance for each month.

What's More

Activity: Which Earns More?

Directions: Figure out which of the two options will give a higher commission by solving each item. Show your solutions.

1. Jenny was able to sell Php 120,000 of the company products in one week.
Option A: Receiving a Php 3,000 salary a week plus a 10% on all sales above Php 75,000
Option B: Receiving a 7% commission on all her sales

2. An agent closed a deal amounting to Php 100,000 in one month.
Option A: Receiving a 5% commission on all the sales
Option B: Receiving a 3% commission on the first Php 50,000 sales, then 6% commission on all sales above Php 50,000

3. A salesperson sold an item that costs Php 75,000.
Option A: Receiving a 5% commission on all the sales
Option B: Receiving a 3% commission on the first Php 50,000 sales, then 7.5% commission on all sales above Php 50,000

4. Enzo's sales in one month is Php 98,500.
Option A: Receiving a 10% commission on all sales.
Option B: Receiving a Php 10,000 monthly salary plus a 2% commission on all sales above Php 50,000

5. Lois sold eight television sets that cost Php 45,000 each in a week.
Option A: Receiving a 7.5% commission on all the sales
Option B: Receiving a Php 3,000 weekly salary plus a 5% commission on all sales

What I Can Do

Instructions: Read the problem carefully. Then, complete the table and answer the questions below.

A client made a purchase using her credit card. The total amount due at the month's cut-off date is Php 70,548.75. The minimum required payment is 6.6% of the total amount due. As stated in the agreement form, if the client pays only the minimum required payment, a financing charge of 4.5% of the remaining balance will be carried on every billing period. If there are no credit bill transactions for the next five months and only the minimum required payment is paid, show the total amount due for each of the five months.

Complete the table:

Time (month)	Total Amount Due for the Month	Minimum Required Payment for the Month
0		
1		
2		
3		
4		
5		

1. How much of the payment goes to interest payments?
   _____
   _____

2. How much of the payment goes to the payment of the actual debt?
   _____
   _____

MODULE 10

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Business Mathematics. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of learners. The lessons are arranged to follow the standard sequence of the course.
This module targets the following learning competency:
Solve problems involving interests and commissions (ABM_BM11BS-IIb-14).

After going through this module, you are expected to:

- analyze each problem to identify the unknown and the given information; and
- apply the procedures for finding the interest and commission.

Lesson 10	Interest and Commission Problems

In the previous modules, you studied how to compute interest and commission. Considering these topics is essential because you can use it to protect the fruit of your labor or to grow your income in the future. Hence, you must take time to study how to deal with problems involving these, as this can help you develop your skill in calculating. In this module, you will solve real-life problems involving these topics: interest and commission.

What is It

It is undeniable that people need money to buy necessities in life. This is one of the reasons why many people work hard just to earn it. However, it will only come to waste if the person using it does not know how to manage it well. Thus, making wise financial decisions is essential. You need to weigh things before entering in to any contract or agreement.

Interest and commissions deal with money. To make right decisions, you need to know how to compute and solve problems related to these.

Recall the formula for simple interest:

$I = PRT$

where I = Interest charged
P = Principal
R = Rate of interest
T = Term or Interest Period

Take note that R and T must be of the same unit. For instance, if R is an annual or yearly interest rate, then T must also be written in years.

Express the following periods in terms of years.

Time	Term in Years
73 days	$\frac{73}{365}$ or 0.2 year
120 days	
6 months	$\frac{6}{12}$ or 0.5 year
10 months	
15 weeks	
36 weeks	

Knowing the basic concepts in fraction, decimal, and percent is necessary for dealing with problems involving interest and commissions.

Example 1:

A lending company offers a simple interest rate of 7.25% per annum. How much is the interest in three years if a client loan Php250,000 for business purposes?

Solution:

Steps	**Solutions**
Identify the given.	P = Php250,000 R = 7.25% or 0.0725 per year T = 3 years *simple interest *(Note that R and T are both expressed in years.)*
Solve for the interest in three years.	I = PRT I = (250,000) (.0725)(3) I = 54,375
State the final answer.	The interest in three years when a client loans Php250,000 is Php54,375.

Example 2: Rose invested Php35,000 in a business with an agreement that she will earn a 10% simple interest per year. How much does she earn in 9 months?

Solution:

Steps	Solutions
Identify the given.	P = Php35,000 R = 10% or 0.10 per year T = 9 months *simple interest *(Note that R and T are not of the same unit. To express months in terms of years, you have to divide 9 by the total number of months in a year.)*
Solve for the interest in three years.	I = PRT I = $(35,000)(.10)\left(\frac{9}{12}\right)$ I = 2,625
State the final answer.	Rose earns Php2,625 in 9 months.

Example 3: Enzo works in a company that offers a 15% commission for every sold item. In one month, he was able to sell items worth Php358,500. He then decided to invest the commission he received to a lending company that offers a 2.75% simple interest per annum. If he decided to withdraw all the amounts in nine (9) months, how much is the total money he received?

Solution:

Steps	Solutions
Identify the given.	Commission Rate = 15% Total Sales = Php358,000
Solve for the commission earned.	C = 0.15(358,000) C = 53,700
Enzo invested this to a business. Compute the interest earned in 9 months.	I = PRT I = $53700(.0275)(\frac{9}{12})$ I = 53700(.0275)(0.75) I = 1107.56 Add the interest to the principal amount he invested. 53700 + 1107.56 = 57807.56
Make a concluding statement.	Enzo received Php57,807.56 from the lending company.

Example 4: A client invested his Php50,000 to a business that offers a simple interest rate. If this client gained a Php2,500 interest in 2 years, what is the interest rate?

Solution:

Steps	Solutions
Identify the given.	Principal = Php50,000 Interest = Php2,500 Term = 2 years
Find the interest rate using the formula I = PRT.	I = PRT

• Substitute the given to the formula, then solve for R.	2500 = (50,000)(R)(2) 2500 = 100,000R $\frac{2,500}{100,000} = \frac{100,000\,R}{100,000}$ 0.025 = R (in decimal)
Multiply the result by 100, then add % symbol.	R = 2.5%
Make a concluding statement.	The interest rate of the business where the client invested Php50,000 is 2.5%.

Example 5: A salesperson received Php2,925 as a commission for selling products worth Php58,500. What is the rate of commission?

Solution:

Steps	Solutions
Identify the given.	Commission = Php2,925 Total Sales = Php58,500
Solve for the commission rate. • Recall that the formula for commission is: Commission = Total Sales x Commission Rate • Substitute the given to the formula. • Divide both sides by the numerical coefficient of R	• C = TS*R • 2925 = 58500*R • $\frac{2925}{58500} = \frac{58500R}{58500}$ .05 = R R = 5%
Make a concluding statement.	The commission rate that was given to the salesman is 5%

What's More

Activity: Fill Me Out!

Instructions: Complete the table by applying the concepts that you have learned from interests and commissions.

A. Interest

	Simple Interest	Principal	Rate	Time
1		Php 12,750	0.7%	48 months
2	Php 9,000	Php 150,000		3 years
3		Php 4,000	20%	6 years
4	Php 5,887.50	Php 78,500	3.75%	
5	Php 15,000		1.5%	1 year

B. Commission

	Commission	Rate	Sales
6		2.5%	Php 350,000
7	Php 600		Php 20,000
8	Php 10,000	10%	
9	Php 4,400		Php 80,000
10		6%	Php 400,000

What I Can Do

Wait, Let's Weigh!

Directions: Read the problem below and weigh the options before answering the questions.

Phoebe is a supplier agent. Every month, she receives Php12,000 plus a 15% commission on all her sales above Php150,000. Her usual monthly sales range from Php250,000 to Php350,000 because she's good at marketing the products. For some reason, her sales per month never gets below Php250,000 and never gets higher than Php350,000. One day, another supplier of the same business as her current supplier came to her with a 12% commission rate offered. If you were Phoebe and you are after a higher compensation, which of the two suppliers will you choose: the current or the new one? Explain.

Answer:

MODULE 11

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Business Mathematics. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of learners. The lessons are arranged to follow the standard sequence of the course.

This module targets the following learning competencies:
1. Define salary, wage, income, benefits (ABM_BM11SW-IIc-1).
2. Compute gross and net earnings (ABM_BM11SW-IId2).

After going through this module, you are expected to:
- define salary and wage;
- differentiate between salary and wage;
- distinguish between salary earners and wage earners;
- define income and benefits; and
- compute gross and net earnings.

Lesson 11 Salaries and Wages

A business will not operate without a workforce. It may consist of one (the owner) or more people working together to achieve a common goal. In return, these people have the right to receive compensation for the time and service they employ in a company or business entity. In this module, you will identify different ways of how these people are being compensated. You will also study how to compute their earnings.

What is It

Compensation is an essential aspect of managing a business. It is a form of transaction where an employee receives a payment in exchange for his or her service to an employer. Compensation can be in direct forms (e.g., basic pay and incentive pay) or indirect forms (e.g., vacation pay and health insurance).

There are varied ways of compensating for a service. One of these is by giving salaries and wages. Salary is a regular payment quoted either monthly or annually. It is fixed and does not change regardless of the number of hours an individual worked for the company. Meaning there is no extra pay if they work beyond their working hours. Those who receive a salary as a form of compensation are the salary earners. On the other hand, those who receive wages are called wage earners. In contrast to salary, the wage is calculated based on the number of hours or days an employee worked. Thus, working for extra hours means receiving extra payment.

Examples:

Salary Earner	Wage Earner
A manager who receives Php 30,000 every month.	A tutor who is paid Php150 per hour.

➢ The daily wage of a salary earner ***depends on a fixed amount*** per month. In this example, the Php 30,000 earning will not change whether there are 28 days or 30 days in a month. If there are 28 days in a month, the daily wage is Php 1,071.43. If there are 30 days, then the daily wage becomes Php 1,000.	➢ The compensation of a wage earner in a month depends on the number of hours or days he or she worked in a company. In the example, if the tutor coaches a child for 40 hours in a month, then her pay in a month for that job would be Php6,000. If she tutors for 45 hours, then her compensation for that month would be Php6,750.

Again, an employee is a salary earner if his or her compensation is not based on an hourly rate but on a monthly or annual rate. Conversely, an employee is a wage earner if his or her earnings are computed based on the number of hours or days he or she worked in a company. Hence, when you calculate the daily or hourly rate of an employee, the computed amount is a wage. You do not say "How much is your daily salary?" and not even "How much is your salary per hour?"

Example: A saleslady earns a salary of Php 11,520 per month. If she works 8 hours a day for working 24 days in a month, how much is her daily and hourly wage?

(Solution: 11,520/24 days = 480 per day; 480/8 hours = 60 per hour)

You should know how to compute the daily rate or hourly rate of a person even if he or she is earning a salary. Why? It is because an amount for being late or absent should be deducted from an employee's salary at the end of the month.

A. Income and Benefits

Income is the money earned by a person in a particular period. Some may earn it from doing a job, and some by investing in entities like businesses, stocks, or banks. The source of income also differs depending on the nature of the work of an individual. For instance, a farmer earns an income by offering goods such as crops, fruits, and vegetables. The most common sources of income are salaries and wages.

Another way to compensate for a person's service is by giving them benefits. It is not the same with income because some benefits are intended for a specific purpose like clothing and medical health costs. It is given on top of the basic salary or wage. It could be in the form of cash (i.e., 13th-month pay, food allowance, and clothing allowance) or in-kind (i.e., Pag-IBIG, Philhealth, and Life Insurance).

B. Gross and Net Earnings

Gross earnings (or gross income) refer to the total amount an individual receives before taking out any deductions. There are two types of deductions. These are the voluntary (i.e., life insurance and loans) and mandatory deductions (i.e., income tax and SSS/GSIS). It includes the basic pay, overtime pay, bonuses, commissions, and other forms of earnings you receive from your employer. **Net earnings** (or net income), on the other hand, refer to the amount an employee takes home on payday. It is the remaining amount after subtracting all the voluntary and mandatory deductions.

Example 1: How much is the monthly gross income if the annual gross income is Php 300,000?

Solution: Monthly Gross Income = Annual gross income ÷ 12 months

Monthly Gross Income = Php 300,000 ÷ 12 = **Php 25,000**

Example 2: A salesperson gets a basic pay of Php 20,000 a month, a food and transportation allowance of Php 1,500 a month, and a commission of 3% of all his sales is a month. How much is his gross income if he was able to sell Php 120,000 in a month?

Solution:

Basic Pay	Php 20,000.00
Food and transportation allowance	1,500.00
Commission (3% of Php 120,000.00)	3,600.00
Total Income (Gross Pay)	**Php 25,100.00**

The salesperson received Php 25,100.00 in that particular month.

Example 3: Suppose that the monthly gross income of an employee is Php 20,000 and she has to pay **annually** Php 3,600 for PhilHealth, Php 1,200 for Pag-IBIG, and Php 9,600 for SSS:

a. How much are her monthly deductions?

b. How much is her monthly net income?

Solution:

a. Compute the monthly deductions by dividing the amount by 12 months. Hence, you will get the following results per month:

PhilHealth	Php 300.00
Pag-IBIG	100.00
SSS	800.00
Total Deductions	**Php 1,200.00**

b. Compute the monthly net income by subtracting the amount of total deduction from her monthly gross income.

Gross Income	Php 20,000.00
Less: Total Deductions	1,200.00
Net Income	**Php 18,800.00**

What's More

Activity: Who Earns the Most?

Instructions: Find out who among the employees earns the most by ranking their names based on their **annual** gross earning. The ranking should be in descending order.

• Rose, who earns Php 345,000 per annum • Lyn, who earns Php 24,000 per month • Jen, who works 40 hours a week at Php 80 per hour	1 _____ 2 _____ 3 _____
• Arnel, who works 56 hours a week at Php 65 per hour • Enzo, who works 5 days a week at Php 1,500 a day • Adolfo, who earns Php 420,000 annually	1 _____ 2 _____ 3 _____
• Employee A, who earns Php 30,000 per month • Employee B, who works 36 hours a week at Php 300 per hour • Employee C, who earns Php 375,000 annually	1 _____ 2 _____ 3 _____

What I Can Do

It's a Payday!

Directions: Select the employee with a higher take home pay based on the given situations. Then, state by how much the difference of their net earnings is.

1. Rico and Rica are both employees of ABC Company. Rico is a supervisor while Rica is a saleslady. Rico's monthly salary is Php 16,800. Aside from this, he also receives a food and transportation allowance of Php 1,500 a month. However, a deduction of Php 3,500 is subtracted from his salary every payday. Rica, on the other hand, earns Php 11,000 a month after all deductions have been made. Who between the two has a higher take-home pay?

   Answer:

2. An instructor gets a basic monthly salary of Php 30,000 and a living allowance of Php 3,000 per month. However, this instructor has to pay monthly Php 5,000 for her loan and Php 3,600 for all the mandatory deductions. On the other hand, a private tutor earns Php250 per hour for tutoring a child. She works 40 hours a month and a deduction of Php 500 for monthly dues is being subtracted from her pay. Who between the two has a higher take-home pay?

   Answer:

MODULE 12

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Business Mathematics. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of learners. The lessons are arranged to follow the standard sequence of the course.

This module targets the following learning competencies:

1. Define each of the benefits given to wage earners (ABM_BM11SW-IIe-5).
2. Distinguish taxable from nontaxable benefits (ABM_BM11SW-IIe-6).
3. Enumerate the standard deductions with the corresponding computation (ABM_BM11SW-e-6).

After going through this module, you are expected to:

- define and enumerate the benefits given to wage earners;
- determine whether a benefit is taxable or nontaxable;
- enumerate the standard deductions with the corresponding computation; and
- calculate gross and net earnings.

Lesson 12 Employee Benefits

The previous module tackled some ways on how to compensate an employee working for the benefit of the company. You also experienced how to calculate his or her earnings. In this module, you will study another type of compensation that may help a business attract dedicated and committed employees – employee benefits. These are an essential part of the company's compensation system to encourage more employees to perform better. When the employee works well, the business has a higher tendency to grow and succeed.

What is It

Success will not be possible without people who will work towards the vision and mission of a company. To sum up, they are the real asset of any business. They do a significant contribution to business undertakings.

Studies have shown that there is a relationship between employee satisfaction and company financial performance (Hyacinth, 2019). What does it mean? If you want to profit, motivate your people. The reason is that people perform at their best when they are motivated. Hence, a business must know how to take care of its people.

One way to motivate people is by offering the right benefits. It is not the only motivating factor, but this can make an employee feel valued. This may result in a higher commitment and outstanding performance.

The benefit is given in addition to salary or wage. This can be in the form of cash or in-kind (or non-cash). Examples of cash benefits are the 13th month pay,

14th month pay, living allowance, and bonuses while SSS/GSIS, Pag-IBIG, and PhilHealth are examples of non-cash benefits.

Employee Benefits

Every company has its way of providing benefits for their employees. For instance, a company may offer a benefit that is different from another company. Though benefits vary from one company to another, some benefits are compulsory and mandated by the law. In our country, the Labor Code of the Philippines known as the Presidential Decree No. 442 protects the rights to benefits of all employees. Hence, all employed individuals should receive the benefits required by the government. Otherwise, an employee may file a case against his or her employer for not abiding with the law.

Mandatory benefits of an employee include the following:

- **Minimum Wage**
  - The minimum wage varies from one region to another. In NCR, the minimum wage (as of the year 2020) is Php 537.00 per day.
- **Overtime Pay**
  - A wage earner who performs work over eight (8) hours a day should receive overtime pay. The computation depends on the time when the overtime work happens. In the next module, you will learn how to compute this.
- **Leave Benefits**
  - This benefit allows an employee to be absent from work for a certain period due to different reasons. The number of days depends on the type of leave benefit. Examples of leave benefits are maternity leave, paternity leave, and solo-parent leave.
- **13th Month Pay**
  - An employee who has worked for at least one (1) month is eligible for a 13th month pay. To compute this, just multiply the monthly basic salary by the employment length then divide it by 12 months. Note that a 13th month pay is not the same as the Christmas Bonus. As the name suggests, a bonus is given according to the generosity of the employers. The company is not required to give this to an employee.
- Mandatory Government Contributions
  - All employees below 60 years old who receive more than Php 1,000 per month are required to contribute to the three social insurance funds in the Philippines:
    - ✓ the Social Security System (SSS) for private employees or the Government Service Insurance System (GSIS) for government employees
    - ✓ the Home Development Mutual Fund (HDMF, known as Pag-IBIG Fund)
    - ✓ the Philippine Health Insurance Corporation (PhilHealth)

B. Taxable and Nontaxable Benefits

Some benefits are not required by the law. We call these "de minimis benefits." While these are not compulsory, the government encourages employers to provide these to their employees. "De minimis benefits" are the small privileges offered to an employee to protect the general well-being of the employees. The good thing about this type of benefit is that it is exempted for income tax purposes provided that it does not exceed Php 90,000. All other allowances regularly earned by the employees are subject to income tax purposes under certain conditions. The updated list of *de minimis benefits* are as follows:

1. Monetized unused vacation leave credits of private employees not exceeding ten (10) days during the year
2. Monetized value of vacation and sick leave credits paid to government official and employees
3. Medical cash allowance to dependents of employees, not exceeding Php 1,500 per employee per semester or Php 250 per month
4. Rice subsidy of Php 2,000 or one (1) sack of 50-kilogram rice per month amounting to not more than Php 2,000
5. Uniform and clothing allowance not exceeding Php 6,000 per annum
6. Actual medical assistance, e.g., medical allowance to cover medical and healthcare needs, annual medical/executive check-up, maternity assistance, and routine consultations, not exceeding Php 10,000 per annum
7. Laundry allowance not exceeding Php 300 per month
8. Employee's achievement awards, e.g., for length of service or safety achievement, which must be in the form of tangible personal property other than cash or gift certificate, with an annual monetary value not exceeding Php 10,000 received by the employee under an established written plan which does not discriminate in favor of highly paid employees
9. Gifts made during Christmas and major anniversary celebrations not exceeding Php 5,000 per employee per annum
10. Daily meal allowance for overtime work not exceeding twenty-five percent (25%) of the basic minimum wage
11. Benefits received by an employee by virtue of a collective bargaining agreement (CBA) and productivity incentive schemes provided that the total annual monetary value received from both CBA and productivity incentive schemes combined do not exceed ten thousand pesos (Php 10,000) per employee per taxable year

Source: https://www.bir.gov.ph/index.php/tax-information/withholding-tax.html#wt09

Based on law, all other benefits that are not included in the list of "de minimis benefits" are taxable. Also, if a company provides more than the indicated limitations, the amount beyond the limit would be taxable. Lastly, the amount more than Php 90,000 tax exemption (including the 13th month pay and other benefits such as productivity and incentive bonuses) will be taxable.

Example: Ms. Jen is an employee of ABC Company. She received the following **annual** compensations and benefits.

Annual Basic Salary	Php 350,000.00
13th Month Pay and bonuses	83,000.00
Rice Subsidy	30,000.00
Clothing Allowance	7,500.00

The next table shows which of the benefits is taxable and nontaxable.

Description	Taxable	Nontaxable
De Minimis Benefits:		
Rice Subsidy (Ceiling amount is Php 2,000 per month15)		Php 24,000.00
	Php 6,000.00	
Clothing Allowance (Ceiling amount is Php 6,000 per year)		Php 6,000.00
	Php 1,500.00	
Other Benefits:		
13th Month Pay and bonuses		Php 83,000.00
Total	Php 7,500.00	Php 113,000.00

C. Mandatory Deductions

Basically, the formula for computing taxable income is:

Taxable Income = (Basic Salary + Additional Pay) – Deductions

where:

Basic Salary = the amount of money an employee receives before any deductions or any additional pay

Additional Pay = refer to the amount that is being added to the basic salary such as overtime pay

Deductions = the amount of money that is subtracted from the gross income.

Some reasons for deductions are:

- tardiness
- absences
- loans
- tax and mandatory contributions

Now, let us talk about the mandatory government deductions from salary.

1. **Income Tax**

   This amount is known as the **withholding tax** because the employer does not give this to the employee. Instead, the employer subtracts this amount from the employee's salary then remits this later to the Bureau of Internal Revenue (BIR).

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REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF FINANCE
BUREAU OF INTERNAL REVENUE

ANNEX A

REVISED WITHHOLDING TAX TABLE
Effective January 1, 2018 to December 31, 2022

DAILY	1	2	3	4	5	6
Compensation Level (CL)	685 and below	685	1,096	2,192	5,479	21,918
Prescribed Minimum Withholding Tax	0.00	0.00 + 20% over CL	82.19 + 25% over CL	356.16 + 30% over CL	1,342.47 + 32% over CL	6,602.74 + 35% over CL
WEEKLY	1	2	3	4	5	6
Compensation Level (CL)	4,808 and below	4,808	7,692	15,385	38,462	153,846
Prescribed Minimum Withholding Tax	0.00	0.00 + 20% over CL	576.92 + 25% over CL	2,500.00 + 30% over CL	9,423.08 + 32% over CL	46,346.15 + 35% over CL
SEMI-MONTHLY	1	2	3	4	5	6
Compensation Level (CL)	10,417 and below	10,417	16,667	33,333	83,333	333,333
Prescribed Minimum Withholding Tax	0.00	0.00 + 20% over CL	1,250.00 + 25% over CL	5,416.67 + 30% over CL	20,416.67 + 32% over CL	100,416.67 + 35% over CL
MONTHLY	1	2	3	4	5	6
Compensation Level (CL)	20,833 and below	20,833	33,333	66,667	166,667	666,667
Prescribed Minimum Withholding Tax	0.00	0.0 + 20% over CL	2,500.00 + 25% over CL	10,833.33 + 30% over CL	40,833.33 + 32% over CL	200,833.33 +35% over CL

Source: https://birtaxcalculator.com/tax_tables.php

2. **SSS/GSIS Contribution**

   An employee working in the private sector is required to be an SSS member (or GSIS member if the employee works in a public institution). SSS/GSIS is an insurance program for all employees. However, every member needs to contribute a percentage of their salary.

EMPLOYED MEMBERS

RANGE OF COMPENSATION	EMPLOYED							
	MONTHLY SALARY CREDIT	SS CONTRIBUTION			EC CONTRIBUTION	TOTAL CONTRIBUTION		
		ER	EE	TOTAL	ER	ER	EE	TOTAL
BELOW 2,250	2,000	160.00	80.00	240.00	10.00	170.00	80.00	250.00
2,250 – 2,749.99	2,500	200.00	100.00	300.00	10.00	210.00	100.00	310.00
2,750 – 3,249.99	3,000	240.00	120.00	360.00	10.00	250.00	120.00	370.00
3,250 – 3,749.99	3,500	280.00	140.00	420.00	10.00	290.00	140.00	430.00
3,750 – 4,249.99	4,000	320.00	160.00	480.00	10.00	330.00	160.00	490.00
4,250 – 4,749.99	4,500	360.00	180.00	540.00	10.00	370.00	180.00	550.00
4,750 – 5,249.99	5,000	400.00	200.00	600.00	10.00	410.00	200.00	610.00
5,250 – 5,749.99	5,500	440.00	220.00	660.00	10.00	450.00	220.00	670.00
5,750 – 6,249.99	6,000	480.00	240.00	720.00	10.00	490.00	240.00	730.00
6,250 – 6,749.99	6,500	520.00	260.00	780.00	10.00	530.00	260.00	790.00
6,750 – 7,249.99	7,000	560.00	280.00	840.00	10.00	570.00	280.00	850.00
7,250 – 7,749.99	7,500	600.00	300.00	900.00	10.00	610.00	300.00	910.00
7,750 – 8,249.99	8,000	640.00	320.00	960.00	10.00	650.00	320.00	970.00
8,250 – 8,749.99	8,500	680.00	340.00	1,020.00	10.00	690.00	340.00	1,030.00
8,750 – 9,249.99	9,000	720.00	360.00	1,080.00	10.00	730.00	360.00	1,090.00
9,250 – 9,749.99	9,500	760.00	380.00	1,140.00	10.00	770.00	380.00	1,150.00
9,750 – 10,249.99	10,000	800.00	400.00	1,200.00	10.00	810.00	400.00	1,210.00
10,250 – 10,749.99	10,500	840.00	420.00	1,260.00	10.00	850.00	420.00	1,270.00
10,750 – 11,249.99	11,000	880.00	440.00	1,320.00	10.00	890.00	440.00	1,330.00
11,250 – 11,749.99	11,500	920.00	460.00	1,380.00	10.00	930.00	460.00	1,390.00
11,750 – 12,249.99	12,000	960.00	480.00	1,440.00	10.00	970.00	480.00	1,450.00
12,250 – 12,749.99	12,500	1,000.00	500.00	1,500.00	10.00	1,010.00	500.00	1,510.00
12,750 – 13,249.99	13,000	1,040.00	520.00	1,560.00	10.00	1,050.00	520.00	1,570.00
13,250 – 13,749.99	13,500	1,080.00	540.00	1,620.00	10.00	1,090.00	540.00	1,630.00
13,750 – 14,249.99	14,000	1,120.00	560.00	1,680.00	10.00	1,130.00	560.00	1,690.00
14,250 – 14,749.99	14,500	1,160.00	580.00	1,740.00	10.00	1,170.00	580.00	1,750.00
14,750 – 15,249.99	15,000	1,200.00	600.00	1,800.00	30.00	1,230.00	600.00	1,830.00
15,250 – 15,749.99	15,500	1,240.00	620.00	1,860.00	30.00	1,270.00	620.00	1,890.00
15,750 – 16,249.99	16,000	1,280.00	640.00	1,920.00	30.00	1,310.00	640.00	1,950.00
16,250 – 16,749.99	16,500	1,320.00	660.00	1,980.00	30.00	1,350.00	660.00	2,010.00
16,750 – 17,249.99	17,000	1,360.00	680.00	2,040.00	30.00	1,390.00	680.00	2,070.00
17,250 – 17,749.99	17,500	1,400.00	700.00	2,100.00	30.00	1,430.00	700.00	2,130.00
17,750 – 18,249.99	18,000	1,440.00	720.00	2,160.00	30.00	1,470.00	720.00	2,190.00
18,250 – 18,749.99	18,500	1,480.00	740.00	2,220.00	30.00	1,510.00	740.00	2,250.00
18,750 – 19,249.99	19,000	1,520.00	760.00	2,280.00	30.00	1,550.00	760.00	2,310.00
19,250 – 19,749.99	19,500	1,560.00	780.00	2,340.00	30.00	1,590.00	780.00	2,370.00
19,750 and above	20,000	1,600.00	800.00	2,400.00	30.00	1,630.00	800.00	2,430.00

Source: http://sssanswers.com/sss-contribution-table-2018/new-sss-contribution-table-2019-employed-members/

3. **PhilHealth Contribution**

   The contribution intends to provide health insurance to employees. Just multiply the basic salary by the premium rate. Half of the result is shouldered by the employer and the other half by the employee. Both the employee and employers contribute **equal** shares for PhilHealth.

Year	Monthly Basic Salary	Premium Rate	Monthly Premium
2019	P10,000.00	2.75%	P275.00
	P10,000.01 to P49,999.99		P275.00 to P1,375.00
	P50,000.00		P1,375.00
2020	P10,000.00	3.00%	P300.00
	P10,000.01 to P59,999.99		P300.00 to P1,800.00
	P60,000.00		P1,800.00
2021	P10,000.00	3.50%	P350.00
	P10,000.01 to P69,999.99		P350.00 to P2,450.00
	P70,000.00		P2,450.00
2022	P10,000.00	4.00%	P400.00
	P10,000.01 to P79,999.99		P400.00 to P3,200.00
	P80,000.00		P3,200.00
2023	P10,000.00	4.50%	P450.00
	P10,000.01 to P89,999.99		P450.00 to P4,050.00
	P90,000.00		P4,050.00
2024 to 2025	P10,000.00	5.00%	P500.00
	P10,000.01 to P99,999.99		P500.00 to P5,000.00
	P100,000.00		P5,000.00

Per PhilHealth Circular No. 2019-0009

Source: https://www.philhealth.gov.ph/news/2019/new_contri.php#gsc.tab=0

4. **Pag-IBIG Contribution**
   Please take note that Pag-IBIG uses a **maximum** monthly compensation of Php 5,000. Thus, the **highest share** is Php 100 per employee and Php 100 per employer.

Pagibig Contribution Table

Monthly Compensation	Percentage of Monthly Compensation	
	Employee Share	Employer Share
P1,500 and below	1%	2%
Over P1,500	2%	2%

Source: https://www.howtoquick.net/2018/01/updated-pag-ibig-contribution-table.html

Sample Problem: Enzo works as a salesman in XYZ Private Company. His monthly gross income is Php 30,000. How much is the taxable income and the net income?

Solution: Enzo's gross income (basic salary + additional pay) is Php 30,000.

(To answer the problem, look at the monthly contribution tables that were presented before arriving to this part.)

Gross Salary		Php 30,000.00
Less SSS Contribution	Php 800.00	
PhilHealth Contribution	450.00	
Pag-IBIG Contribution	100.00	1,350.00
Taxable Income		Php 28,650.00
Less Withholding Tax		1,563.40
Net Income/Net Pay		Php 27,086.60

What's More

Activity: The Greatest Asset

Instructions: Decode the message by answering the following problems. Then, explain the message in three to five sentences. (Omit the Peso sign in the final answer.)

A = the withholding tax if the taxable income is Php 20,000
E = the withholding tax if the taxable income is Php 25,000
I = the income tax if the taxable income is Php 21,000
O = the SSS contribution of an employee whose compensation is Php 15,000
U = the company's share for the SSS contribution of its employee earning Php 20,000
B = the employee's SSS contribution earning below Php 2,250
L = the employee's share to PhilHealth if his salary is Php 10,000
M = the employer's PhilHealth contribution if his employee's monthly basic salary is Php 10,000
P = the employee's Pag-IBIG contribution if she is earning Php 1,500 a month
T = the employee's contribution to Pag-IBIG Fund if the monthly compensation is Php 5,000

	F	Y								V	
33.4			600	1600	80	833.4	350	33.4	833.4		833.4

	H								Y			S
100		0	100	833.4	175	15	350	600		833.4	833.4	

	R		Y			R			S	
0		833.4		600	1600		175	600		100

V									S	S		
	0	350	1600	0	80	350	833.4	0			833.4	100

,

YOU WILL CREATE A HEALTHY WORK ATMOSPHERE AND PROVIDE THEM WITH THE TOOLS AND SUPPORT TO DO THEIR JOBS EFFECTIVELY.

-Brigette Hyacinth

What I Can Do

Directions: Compute the taxable income and the net income of the following employees, assuming their company compensates its employees every month and considering the mandatory government deductions.

1. Employee X with a gross income of Php 40,000

Gross Salary		
Less SSS Contribution		
PhilHealth Contribution		
Pag-IBIG Contribution		
Taxable Income		
Less Withholding Tax		
Net Income/Net Pay		

2. Employee Y with a gross income of Php 35,000

Gross Salary		
Less SSS Contribution		
PhilHealth Contribution		
Pag-IBIG Contribution		
Taxable Income		
Less Withholding Tax		
Net Income/Net Pay		

3. Employee Z with a gross income of Php 30,000

Gross Salary		
Less SSS Contribution		
PhilHealth Contribution		
Pag-IBIG Contribution		
Taxable Income		
Less Withholding Tax		
Net Income/Net Pay		

MODULE 13

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Business Mathematics. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of learners. The lessons are arranged to follow the standard sequence of the course.

This module targets the following learning competencies:

1. Identify the variables needed in the computation of the overtime (ABM_BM11SW- IIe -8).
2. Compute overtime pay (ABM_BM11SW- IIe -9).

After going through this module, you are expected to:

- know your overtime pay rights.
- identify the variables needed to compute the overtime; and
- compute overtime pay.

Lesson 13 Overtime Pay

Employers protect their employees by giving them enough time to rest. It is because an overworked employee cannot perform at its best. However, there are times that employees may be required to work overtime when there is urgent work needed to fulfill that may affect the business status. In this module, you will learn how the Labor Code of the Philippines protects the employees with this matter.

What is It

According to Article 83 of the Labor Code, any employee shall work not exceeding 8 hours a day, excluding at least a one-hour lunch break. Working beyond eight hours a day will entitle an employee to an extra pay called overtime pay (*or premium pay if an employee report to work during a non-working day like rest days and holidays*). Articles 93 and 94 specify the hourly rate and overtime premium an employee should receive depending on the workday he or she reports to work. This benefit applies to all employees except to those listed on this site: https://www.laborlaw.ph/legal/overtime-pay-under-philippine-labor-law

To compute the overtime pay:

Step 1: Divide the daily wage by eight to determine the regular hourly wage.

Step 2: Identify the workday an employee reported to work. Refer to the table below. Note that this table is applicable only for non-night shift.

Workday	**Hourly Rate if an employee will NOT be working for more than eight hours**	**Hourly Rate in excess of eight hours**
Ordinary Day		125%
Rest Day	130%	169%
Special Non-working Day	130%	169%
Special Non-working Day that falls on a Rest Day	150%	195%
Regular Holiday	200%	260%
Regular Holiday that falls on a Rest Day	260%	338%

Step 3: Compute the overtime pay using the variables needed, i.e., the regular hourly wage, overtime rate, and overtime.

Example: Jenny works as a saleslady in a store that sells branded clothes. She works from Monday to Friday, and her pay is Php 537.00 a day.

a. Suppose she reported to work from Monday to Friday with no overtime. How much is her pay in a week if she worked eight hours on a regular Saturday (rest day)?

Step	**Solution**
1. Compute her pay from Monday to Friday.	Since there is no overtime from Monday to Friday, just multiply the daily wage by 5. ➢ 537(5) = 2,685
2. Calculate the overtime pay (premium pay).	• Her overtime happened during her rest day. • The overtime rate is 130%. • Rewrite 130% to 1.30 (or 1.3). ➢ 537(1.3) = 698.10
3. Add the results from steps 1 and 2.	➢ 2,685 + 698.10 = 3,383.10
Therefore, her pay during that week is Php 3,383.10.	

b. Suppose that she worked 10 hours on a regular holiday. How much is her pay on that day?

Step	Solution
1. Compute her hourly wage.	Divide the daily wage by 8. • 537/8 = 67.125
2. Calculate her pay on that day. (*Note that aside from working during a regular holiday, she also worked two hours beyond the required working hours. This means, you will be using two different overtime rates.*) • Regular Holiday Rate (for the first 8 hours) – 200% • Overtime rate during a regular holiday – 260%	Determine the pay for the first 8 hours. • 537(200%) = 1,074 Determine the pay for the hours worked beyond eight hours • 67.125 (260%) (2 hours overtime) = 349.05
3. Add the results in Step 2.	• 1,074 + 349.05 = 1,423.05
Therefore, her pay on that day is Php 1,423.05.	

What's More

Activity: Pay Me Right

Directions: Find the pay you must receive based on the following scenarios.

1. My rest day fell on a special non-working holiday. However, the company manager requested me to report to work on that day because there was a task to finish and a deadline to meet. How much was my pay that day if my daily wage is Php 800?

Solution:

2. Suppose that my daily wage is Php 1,000.00, and I worked for 9.5 hours on a normal day. How much is my total pay on that day?

Solution:

3. My daily wage is Php 537.00. If I work on Christmas Day, how much pay will I receive on that day?

Solution:

What I Can Do

Directions: Complete the table.

Workday	Time In	Time Out	Total Number of Hours Rendered	Overtime (No. of hours worked beyond the regular working hours)	Daily Pay (Including the Overtime Pay)
Monday (Ordinary Day)					
Tuesday (Regular Holiday)					
Wednesday (Ordinary Day					
Thursday (Rest Day)					
Friday (Ordinary Day)					
	Total Weekly Pay				

Note: 12:00-1:00 is Lunch Break (Not included in the number of hours rendered in the company.)

MODULE 14

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Business Mathematics. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of learners. The lessons are arranged to follow the standard sequence of the course.

This module targets the following learning competencies:

1. Use E- spread sheet in the computation of salary and overtime pay (ABM_BM11SW- IIf -10).
2. Present graphical representation of the details or particulars of the salary. (ABM_BM11SW- IIf -13).

After going through this module, you are expected to:

- identify the important parts of electronic spreadsheet needed in the computation of salary and overtime pay.
- compute the salary and overtime pay using e-spreadsheet; and
- show graphical representation of the details or particulars of the salary using tables.

Lesson 14 E-spreadsheet in Computing Salary and Overtime Pay

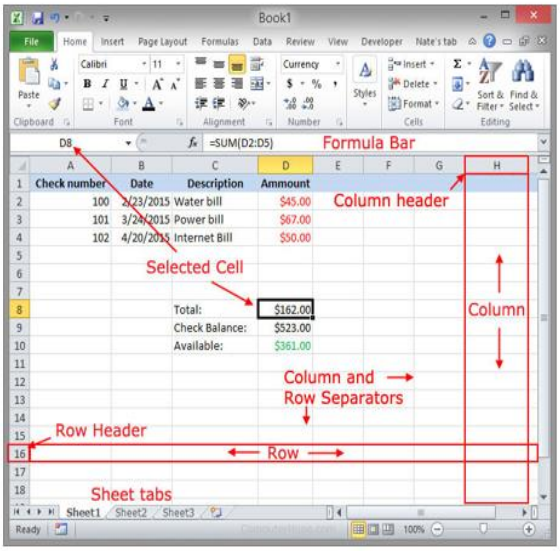
All employees have the right to receive what is due to them. Hence, it is necessary to ensure that they get the right compensation. However, computing salary and overtime pay are quite complicated. In this module, you will learn how to use a program that can help you greatly in doing this task.

What is It

During the old days, people assigned in the payroll used a paper spreadsheet (or ledger paper) to compute salary and overtime pays. It is a form of a table with many rows and columns. These rows and columns intersect and form a rectangular box where the data is being written manually. As the technology progresses, some people were able to invent and develop apps that will help ease the burden of computing. Now, people use an electronic version of a spreadsheet.

If you search on the internet, you will find different spreadsheet programs. Some of these are Google Sheets, Open Office, Microsoft Excel, and many more. Among these, Microsoft Excel is the most widely used.

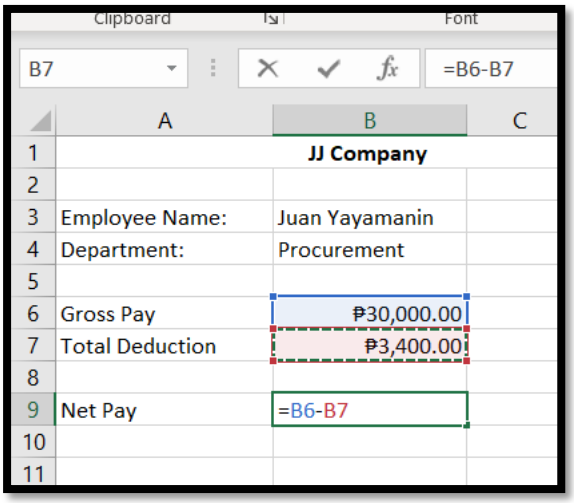
Explore the different parts of a Microsoft Excel spreadsheet.



Source: https://www.computerhope.com/jargon/s/spreadsheet.htm

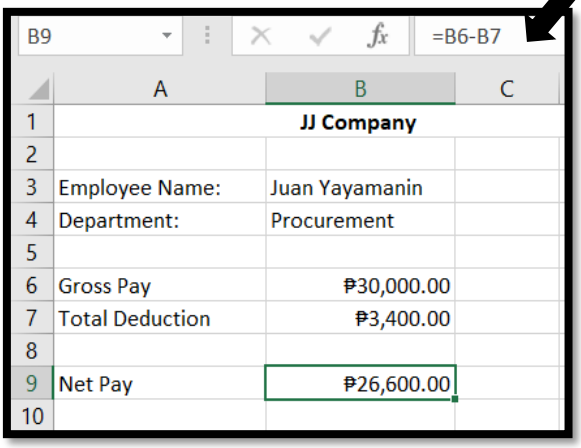
To enter data in a spreadsheet, do the following steps:

1. Choose the cell where you want to enter the data by clicking on it.
2. Encode the data. You will notice that the data you entered on the selected cell is also reflected on the Formula Bar. The name of the selected cell is in the upper left corner beside the formula bar. In the figure above, the Cell Name is D8. It is the intersection of Column D, Row 8.
3. To compute, you need to apply the mathematical principles you have learned previously. For example, subtract the total deduction from the gross pay to calculate the net pay.

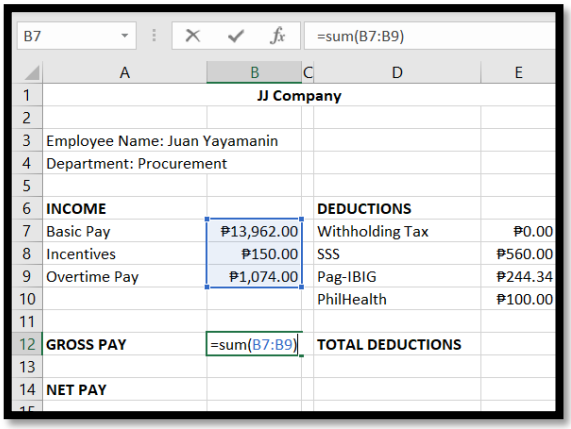


- Start by encoding a formula in the cell. Take note that you should always start with the = sign before encoding any formula.
- Then, type the Cell Name (or click the cell that you want to include in the formula) instead of encoding 30,000 and 3,400. It is to get accurate results even if you change the values you encoded in B6 and B7.

Formula Bar

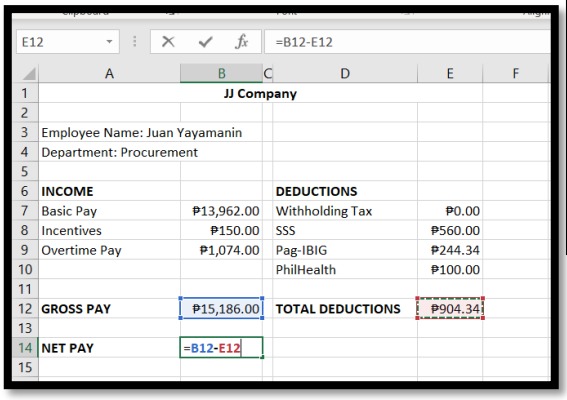


- Click ENTER after you have encoded the formula. You can now see the net pay amount in the cell.
- Since you encoded a formula for net pay, the result here is dependent on B6 and B7. If you modify the amount in B6 or B7, the net pay amount will automatically be changed.
- Observe that if you look at the formula bar, you will see the formula used.



B7 =sum(B7:B9)

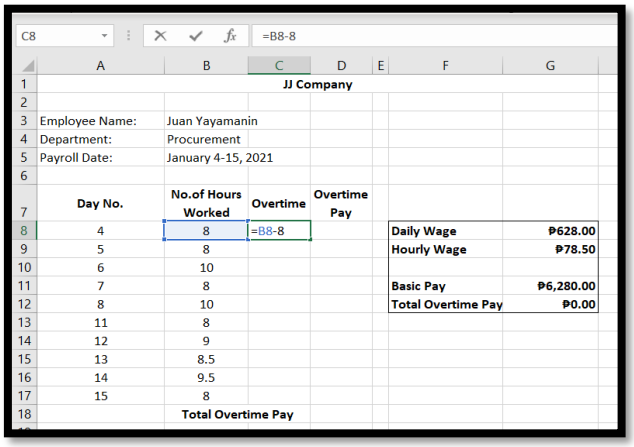
	A	B	C	D	E
1		JJ Company			
2					
3	Employee Name: Juan Yayamanin				
4	Department: Procurement				
5					
6	INCOME			DEDUCTIONS	
7	Basic Pay	₱13,962.00		Withholding Tax	₱0.00
8	Incentives	₱150.00		SSS	₱560.00
9	Overtime Pay	₱1,074.00		Pag-IBIG	₱244.34
10				PhilHealth	₱100.00
11					
12	GROSS PAY	=sum(B7:B9)		TOTAL DEDUCTIONS	
13					
14	NET PAY				



E12 =B12-E12

	A	B	C	D	E	F
1		JJ Company				
2						
3	Employee Name: Juan Yayamanin					
4	Department: Procurement					
5						
6	INCOME			DEDUCTIONS		
7	Basic Pay	₱13,962.00		Withholding Tax	₱0.00	
8	Incentives	₱150.00		SSS	₱560.00	
9	Overtime Pay	₱1,074.00		Pag-IBIG	₱244.34	
10				PhilHealth	₱100.00	
11						
12	GROSS PAY	₱15,186.00		TOTAL DEDUCTIONS	₱904.34	
13						
14	NET PAY	=B12-E12				
15						

- If you want to get the sum of many data, you may encode the = sign, type SUM, then open parenthesis, highlight the cell that you want to add, close parenthesis, then ENTER.
- Remember that to compute the net pay, you need to get the difference between the gross pay and the total deductions.
- To calculate the net pay, click the = sign, click the cell where the **gross pay** amount is located, press the **minus** sign using your keyboard, click the cell where the amount of **total deductions** is located, and the press ENTER.



C8 =B8-8

	A	B	C	D	E	F	G
1			JJ Company				
2							
3	Employee Name:	Juan Yayamanin					
4	Department:	Procurement					
5	Payroll Date:	January 4-15, 2021					
6							
7	Day No.	No.of Hours Worked	Overtime	Overtime Pay			
8	4	8	=B8-8			Daily Wage	₱628.00
9	5	8				Hourly Wage	₱78.50
10	6	10					
11	7	8				Basic Pay	₱6,280.00
12	8	10				Total Overtime Pay	₱0.00
13	11	8					
14	12	9					
15	13	8.5					
16	14	9.5					
17	15	8					
18		Total Overtime Pay					

- To compute the overtime pay, type in the cell after the heading overtime the following:
  - =B8-8
- Fill the cells with the formula by dragging down the lower right corner of the cell and stop at row 17.
- To determine the OT pay per row, encode the following on the first cell after the OT Pay column:
  - =G9*1.25*C8
- Drag down the formula to fill in the cells below.
  - ❖ Note that the Cell Names B8, G9, and C8 may vary from your work because it depends on where you typed the variables needed in the computation.

What's More

Activity: Your Work is Not in Vain

Directions: Suppose an employee, who is earning Php 600.00 a day, receives weekly pay from a company. Determine the total amount of pay by completing the needed data using MS Excel.

Business Name:

Name of Employee:

Workday	No. of Hours Worked	Overtime	Overtime Pay
Monday*	10		
Tuesday*	8.5		
Wednesday*	9		
Thursday*	10		
Friday*	8		

*regular/normal non-night shift days

Daily Wage	600.00
Hourly Wage	
Basic Pay	
Total Overtime Pay	
Gross Pay	

What I Can Do

Instructions:

1. Assume that you have a small business with ten employees. Each employee earns Php 600.00 a day.
2. Think of a business name, an employee name, a department, and a payroll date.
3. Complete the table below by utilizing MS Excel. Use a formula to compute the overtime, overtime pay, total overtime pay, hourly wage, basic pay, and gross pay. (*Note: You may present the table differently, as long as the required data are there.*)

2	Business Name:					
3	Employee Name:					
4	Department:					
5	Payroll Date:					
6						
7	Day No.	No.of Hours Worked	Overtime	Overtime Pay		
8	4	9			Daily Wage	₱600.00
9	5	8			Hourly Wage	
10	6	8				
11	7	8			Basic Pay	
12	8	9.5			Total Overtime Pay	
13	11	8			Gross Pay	
14	12	8.5				
15	13	10				
16	14	9.5				
17	15	8				
18		Total Overtime Pay				
19						
20	*All workdays are regular/normal non-night shifts					

RUBRIC

CRITERIA	4	3	2	1
Completeness	All the required information are filled out	One required information is missing	2-3 required information are missing	More than 3 required information are missing
Mathematical Concept	All computations are correct	1-2 computations are incorrect	3-4 computations are incorrect	More than 4 incorrect computations
Adherence	All instructions were followed	One instruction was not followed	2-3 instructions were not followed	More than 3 instructions were not followed
Presentation	All data are clear and presentable	Most of the data are clear and presentable	Most data are unclear	Needs improvement

MODULE 15

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Business Mathematics. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of learners. The lessons are arranged to follow the standard sequence of the course.

This module targets the following learning competencies:

1. Compare the forms (textual, tabular, and graphical) of business data (ABM_BM11PAD-IIg-2).
2. Describe the different kinds of graphs and its essential parts for data presentation.
3. Give a set of business data; identify the graphs to be used (ABM_BM11PAD-IIi-8).
4. Draw the graph/table to present the data (ABM_BM11PAD-IIi-9).

Lesson 15 — Presentation of Data

Business data serve as a basis for making decisions. However, these will only be mere data if not organized and presented well. Thus, knowing how to show data is necessary, especially if you want to attract investors. In this module, you will learn more about this.

What is It

Presentation of data is about organizing the information to help the readers (or viewers) understand the matter easier. There are different ways to do it. You may show it using texts, tables, or graphs.

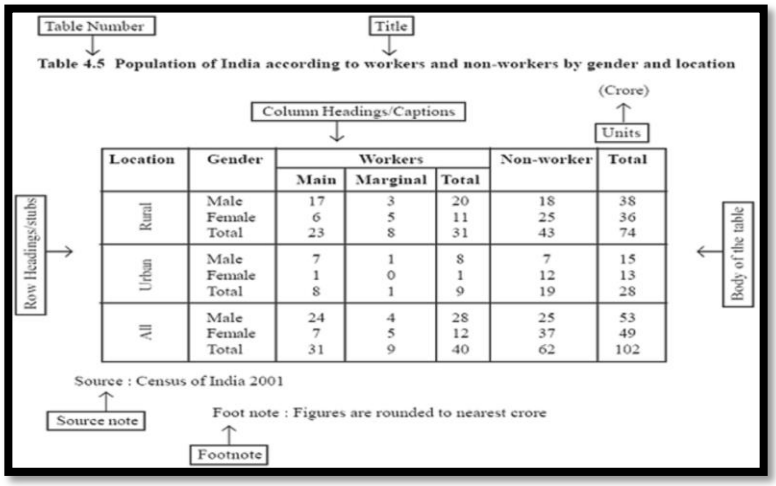
Textual Form

- the data are presented in words
- usually in paragraph form
- used to explain or interpret significant features of the data
- used if the data that you want to show is not very large

Example:
In 2020, five hundred out of seven hundred fifty workers of a factory lost their job due to an outbreak in the city. Forty-five percent of them were female.

Tabular Form

- the data are presented using tables
- easy to read and construct
- can be used even with a large amount of data because it helps simplify complex data
- can be used even if you are presenting information with different units



Location	Gender	Workers			Non-worker	Total
		Main	Marginal	Total		
Rural	Male	17	3	20	18	38
	Female	6	5	11	25	36
	Total	23	8	31	43	74
Urban	Male	7	1	8	7	15
	Female	1	0	1	12	13
	Total	8	1	9	19	28
All	Male	24	4	28	25	53
	Female	7	5	12	37	49
	Total	31	9	40	62	102

Source: https://studyofplant.blogspot.com/2016/08/biostatistics-

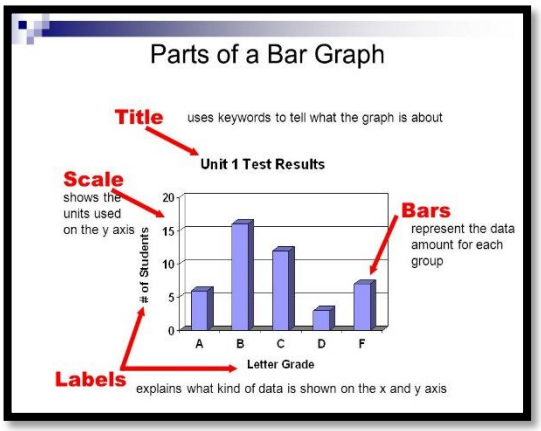
Graphical Form

- the data are presented using graphs
- easy to understand even if the reader (or viewer) is not good in language or numbers
- helps simplify complex data

The common types of graphs are bar chart, pie chart, and line graph.

Bar Chart

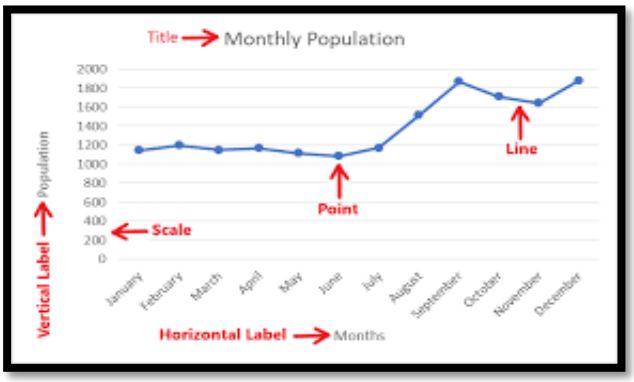
- uses bars of different lengths but with equal widths
- used when the order of categories is not important
- usually used to compare values across a few categories



Source: https://thetablebar.blogspot.com/2018/02/bar-graph-parts.html

Line Graph

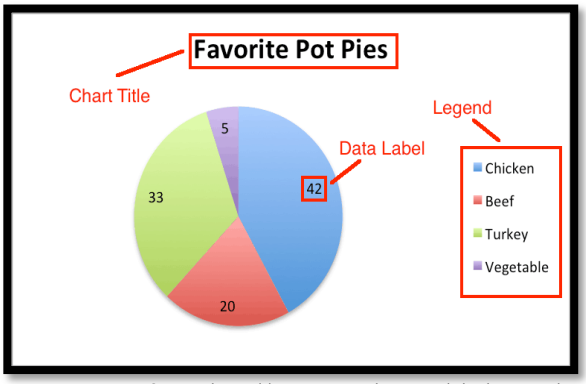
- uses line to present data
- used to show trends over time
- used when the order is important



Source: https://www.formpl.us/resources/graph-chart/line/

Pie Chart

- uses pie slices
- used to show parts of a whole



Source: https://www.smartsheet.com/pie-chart-excel

It is also good to support tabular or graphical presentations with text. However, you don't have to go into detail anymore. When you add text to a table or graph, emphasize only the information you want to highlight. There is no need to write everything that is already in the table or graph.

What's More

Directions: Write the correct answer in the space provided before the number.

_____ 1. Which type of graph is appropriate to use if you want to compare the most popular shoe brands in your barangay?

_____ 2. You planted a mango seed, and you want to measure its growth for ten weeks. What type of graph is best to use?

_____ 3. To show how you allocate your expenses in business, which type of graph is the most appropriate to use?

_____ 4. Which form of data presentation is best to use if the data you want to present is very few?
_____ 5. Your boss wants you to present a detailed quarterly report. What form of data presentation will you use?

What I Can Do

Instructions:

1. Assume that you have a business.
2. Name your business.
3. Present the daily sales of your business from Monday to Friday using tabular, textual, and graphical forms. Use the most appropriate graph.

Rubric:

CRITERIA	4	3	2	1
Completeness	All the required information are presented	One required information is missing	2-3 required information are missing	More than 3 required information are missing
Mathematical Concept	All concepts are correct.	1-2 concepts are misrepresented.	3-4 concepts are misrepresented.	More than 4 concepts are misrepresented.
Presentation	All data are clear and presentable	Most of the data are clear and presentable	Most data are unclear	Needs improvement

MODULE 16

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Business Mathematics. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of learners. The lessons are arranged to follow the standard sequence of the course.
This module targets the following learning competencies:

1. Analyze and interpret the data presented in a graph/table (ABM_BM11PAD-IIi-10).
2. Analyze and interprets the data presented in the table using measures of central tendency and variability and tests of significant differences (ABM_BM11PAD-IIh-5)
3. Use software (i.e., MS Excel, SPSS) programs to compute and present graphical representation of business data (ABM_BM11PAD-IIj-11)

Lesson 16 Analysis of Data

After you have presented the data, the next thing is to analyze its meaning. What do these data tell you? In this module, you will learn ways on how to analyze and interpret a set of data.

What is It

One of the most used tools in data analysis is the **measures of central tendency**. It attempts to describe a set of data by determining the center of the distribution using a single value.

Example:

Monthly Sales in Peso (2021)			
25,000	23,500	24,788	21,988
28,435	26,705	24,927	25,682
23,401	22,999	23,755	27,004

If there is one single value that would represent all these data, what would it be? The answer to this question is called a **central tendency**. Generally, the most common measures of central tendency are the mean, median, and mode. Sometimes they are called averages.

The **mean** is determined by adding all the values, and then divide the sum by the number of values in the data set. Among the three measures, the mean is unique and may not be one of the data values.

Example: What is the average salary of office staff in NCR?

Office Staff Salary in NCR (in Peso)

Salary (X)	Frequency (f)	f (X)
11,000	15	165,000
12,000	18	216,000
13,000	17	221,000
14,000	12	168,000
15,000	14	210,000
TOTAL	76	980,000

$$\bar{x} = \frac{980,000}{76} = 12,894.74$$

Median is another form of average. To find this, arrange all the data from smallest to largest (or vice versa). Then get the middle value. If the number of data is even, add the two middle values and divide it by two.

Example: Find the average salary of office staff in NCR using the median.

21,988
22,999
23,401
23,500
23,755
24,788
24,927
25,000
25,682
26,705
27,004
28,435

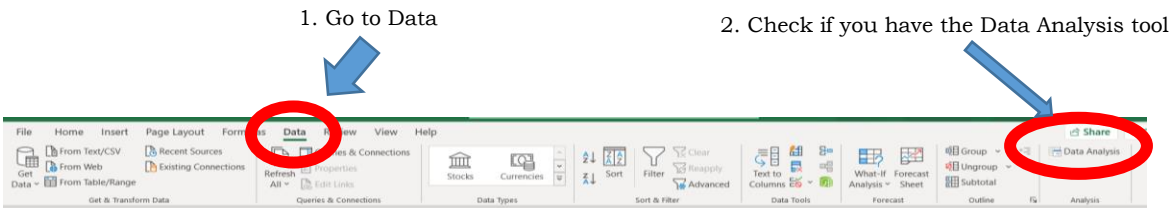
Solution:

- Make sure that all the data are arranged in order.
- Since there are 12 data, get the average of the two middle values:
  Median = (24788 + 24927)/2
  Median = 24,857.50
- Hence, using the median, the average salary is Php24,857.50.

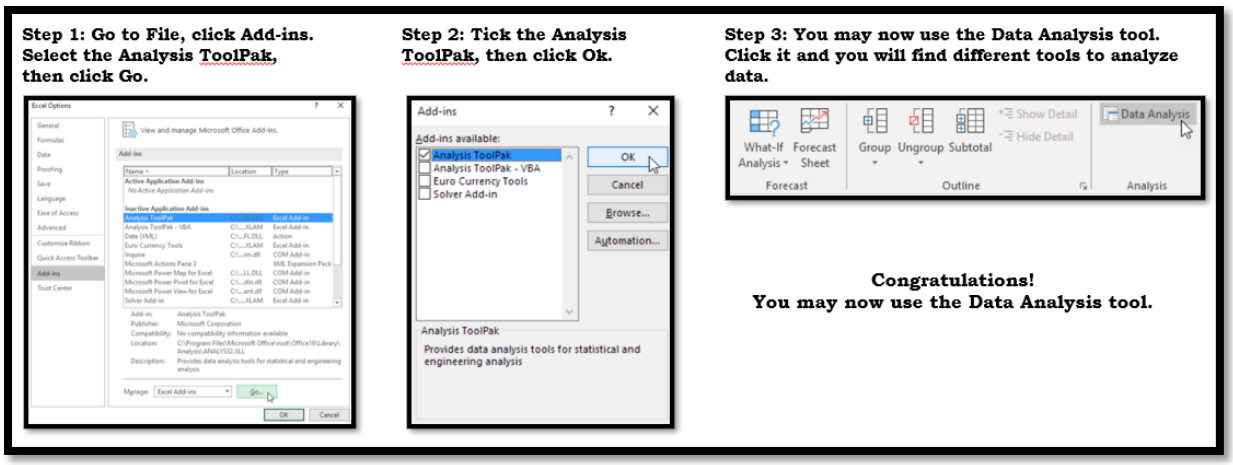
The **mode**, on the other hand, is the easiest to identify. Just look at the data set and determine the value that occurs most often. It represents the highest bar in a bar chart. A data may have one mode (unimodal), two modes (bimodal), three modes (trimodal), more than three modes (multimodal), or no mode at all. In the previous data for office staff salary, no salary appears more than once. Hence, we cannot use the mode in this example.

Good news! You don't have to do this manually. With the aid of computers, you may do all the analysis by using a software program like SPSS and Microsoft Excel. In this module, you will focus on the latter.

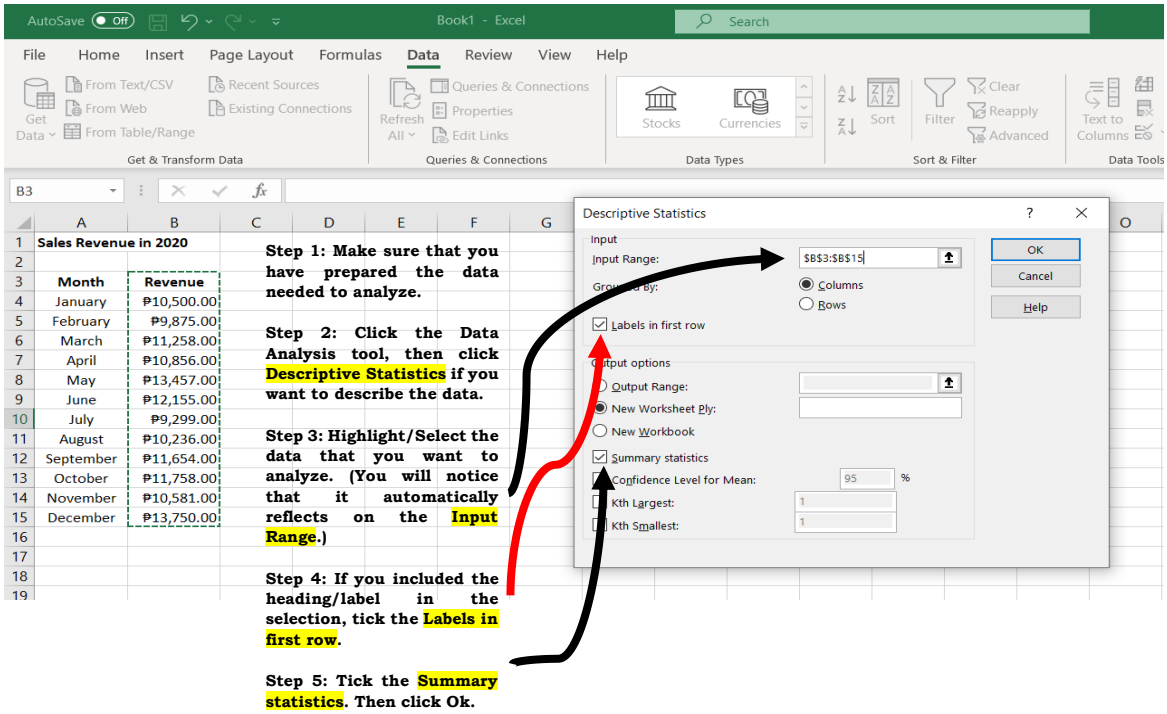
Microsoft Excel has many ways to analyze your data. For now, you will focus on one of the add-in programs that will help you in data analysis – the Analysis ToolPak. To check if you have it in your Excel, do this.



If you don't see the Data Analysis tool, do the following:

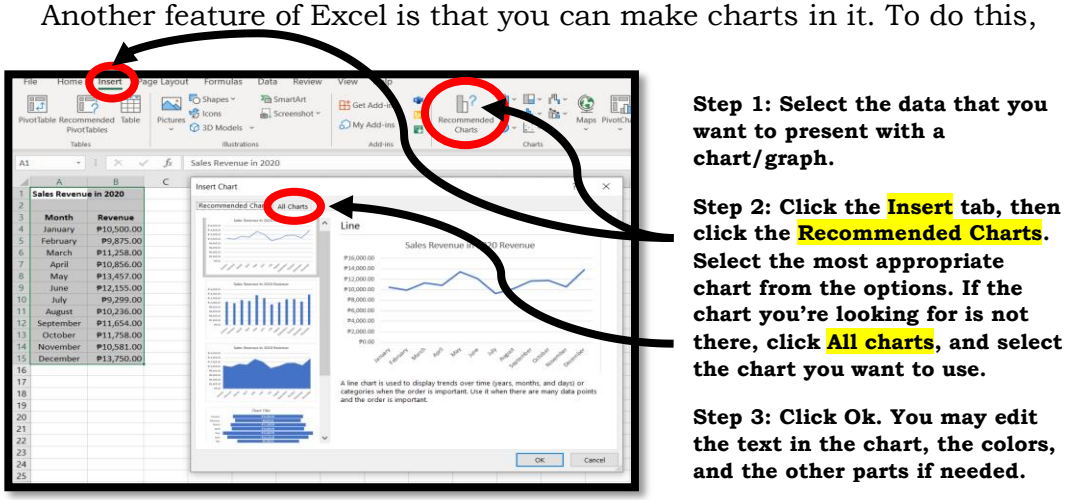


Let's try to do it.



You may explore the tool if you want to measure the variance or test the difference.

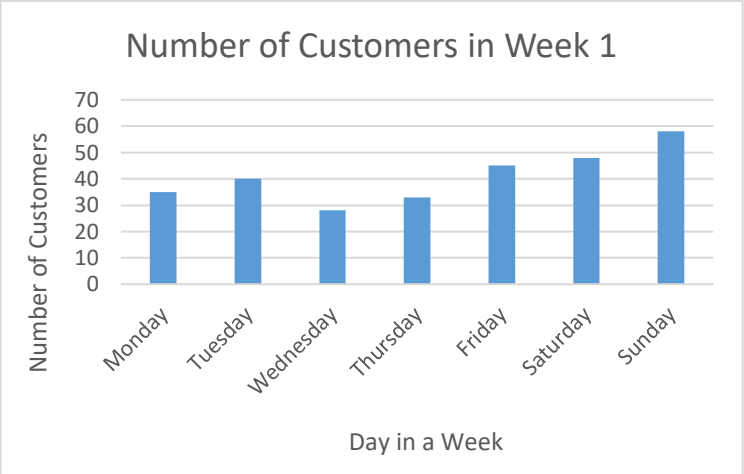
Another feature of Excel is that you can make charts in it. To do this,



After the analysis, the next part is to interpret the result. What can you say about the data? You may look at the trend and find out if there is anything not normal.

What's More

Directions: Answer the questions below the figure.



Questions:

1. What is the title of the graph?

   _____

2. Which day has the least number of customers?

   _____

3. How many customers went to store on Tuesday?

   _____

4. Which day has the most number of customers?

   _____

5. What information can you get on the graph?

   _____

What I Can Do

Instructions:

1. Assume that you have a business.
2. Name your business.
3. Present the monthly sales of your business for the Year 2020 using tabular, textual, and graphical forms. Use the most appropriate graph.
4. Include a descriptive analysis in your textual presentation.
5. Interpret the result.

Rubric:

CRITERIA	4	3	2	1
Completeness	All the required information are presented	One required information is missing	2-3 required information are missing	More than 3 required information are missing
Mathematical Concept	All concepts are correct.	1-2 concepts are misrepresented.	3-4 concepts are misrepresented.	More than 4 concepts are misrepresented.
Presentation	All data are clear and presentable	Most of the data are clear and presentable	Most data are unclear	Needs improvement

Assessment

Directions: Choose the letter of the correct answer. Write your answer on a separate sheet of paper.

Quarter I

1. The more the workers there are, the lesser the number of hours is needed to complete a task. This is an example of what type of proportion?
   A. direct proportion
   B. inverse proportion
   C. partitive proportion
   D. mixed proportion
2. Three out of four children in a school love chocolate products. Represent the fraction of children who love chocolates to decimal form.
   A. 0.34 B. 0.75 C. 3 D. 3.4
4. Rose would like to sell mobile phones she purchased from a supplier. The cost of each phone is Php 3,500.00 each. The operating cost is set at 15% of the cost and she would like to have a 25% profit on the cost of each item. How much should be the selling price?
   A. Php 3,525.00
   B. Php 3,900.00
   C. Php 4,500.00
   D. Php 4,900.00
3. Ramon is 68 years old. As a senior citizen, he is exempted to VAT and is also entitled for a 20% discount on food items. If he orders a meal and the cost of the food is Php 90.00, how much will she pay the cashier?
   A. Php 28.80
   B. Php 61.20
   C. Php 64.29
   D. Php 72.00

For numbers 5-6: 3 pieces of lumpiang sariwa cost Php 120. How much will it cost for 15 pieces?

5. How will you translate this problem into a proportion?
   A. $\frac{3}{120} = \frac{15}{n}$
   B. $\frac{3}{120} = \frac{n}{15}$
   C. $\frac{3}{15} = \frac{n}{120}$
   D. $\frac{3}{n} = \frac{15}{120}$
6. How much will be the cost if you buy 15 pieces of lumpiang sariwa?
   A. Php 240.00
   B. Php 360.00
   C. Php 600.00
   D. Php 720.00
7. PQR Company sells laptops worth ₱24,999 each. Each unit costs 75% of its selling price. What is the rate of margin?
   A. 23.5% B. 25% C. 28.7% D. 75%
8. A worker earns a salary of Php 36,000 a month. Suppose that he works 8 hours a day for a 24-day work month, how much does he get per hour?
   A. Php 150.00
   B. Php 187.50
   C. Php 1,500.00
   D. Php 36,000.00
9. How much is the profit/loss of a business if the revenue is ₱120,500 and the expenses is Php102,000?
   A. Profit; Php 18,500.00
   B. Profit; Php 120,500.00
   C. Loss; Php 18,500.00
   D. Loss; Php 102,000.00

10. Adolfo works in a company that pays Php 2,000 a week and a 12% commission on sales above Php 15,000 in the week. How much will he receive if he sold products worth Php 28,000 in a week.
    A. Php 15,000.00 C. Php 4,650.00
    B. Php 13,000.00 D. Php 3,560.00
11. What is the single equivalent rate if the merchant gives you a trade discount series of 20/10/10?
    A. 43.5% B. 40% C. 35% D. 35.2%

Quarter II

For numbers 12-13: You are planning to buy a condominium unit worth ₱2,000,000. However, your savings is not enough to fully paid the property, so you applied for a mortgage from a bank. The bank required you to make a 10% down payment.

12. How much is the down payment?
    A. Php 200,000.00 C. Php 2,000.00
    B. Php 20,000.00 D. Php 200.00
13. What is the mortgage loan?
    A. Php 2,200,000.00 C. Php 1,800,000.00
    B. Php 2,000,000.00 D. Php 1,600,000.00
14. Which of the following benefits is taxable?
    A. SSS contribution C. Rice subsidy amounting to Php 2,000
    B. Benefits over Php 90,000 D. 13th month pay not exceeding to Php 90,000
15. A car dealer earns a 2.5% commission on sales. If he sells a car for Php 980,000, how much commission will he earn?
    A. Php 24,500.00 C. Php 245,000.00
    B. Php 25,000.00 D. Php 2,450,000.00
16. In entering a formula inside a cell, the first symbol you must write is ___.
    A. > B. < C. = D. ^
17. An employee works as a clerk in ATL Company, a private lending company in a barangay. His monthly gross income is Php 25,000. How much is the employee's share to Pag-IBIG Fund?
    A. Php 100.00 C. Php 200.00
    B. Php 150.00 D. Php 250.00
18. Which chart is appropriate to use if you want to show trends over time?
    A. Line Graph C. Pie Chart
    B. Bar Chart D. Tables
19. A wage earner who earns Php 537.00 a day reported to work during a regular holiday. How much pay will he receive that day?
    A. Php 671.25 C. Php 939.75
    B. Php 805.50 D. Php 1,074.00
20. A wage earner who earns Php 537.00 a day rendered ten hours in work during a regular day. How much pay will he receive that day?
    A. Php 83.91 C. Php 704.81
    B. Php 167.81 D. Php 939.75

Answer Key

What I Know

(Q1)

1. B
2. C
3. B
4. A
5. C
6. D
7. B
8. D
9. A

(Q2)

10. A
11. C
12. D
13. A
14. B
15. B
16. A
17. C
18. D
19. C
20. A

Assessment

(Q1)

1. B
2. B
3. D
4. C
5. A
6. C
7. B
8. B
9. A
10. D
11. D

(Q2)

12. A
13. C
14. B
15. A
16. C
17. A
18. A
19. D
20. C

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